

Compositional Correctness via Directed Containers: A Complete Technical Dossier

MacBeth
(with N. Ghani, R. Langer)
Kodamai

2026-09-24

Draft — Theory pillar assembled; Applications, Formalisation, and Impact bodies in progress.

Abstract

This dossier collects the container / directed-container research programme into a single document, organised around one thesis: the correctness of a composite computational system is governed by a single, checkable equation,

$$\text{a composite is correct} \iff (\mathbf{L}) \wedge [\omega] = 0 \in H^2.$$

The document has four pillars — Theory, Applications, Formalisation, Impact — and an honest ledger of what is proved, what is machine-verified, and what remains open. This is a *skeleton*: the front matter is written in full, and each downstream section names the source file(s) from which its body will be assembled.

Contents

How to read this document	4
1 Overview: One Thesis	5
1.1 The organizing equation	5
1.2 Why this is a foundation and not a slogan	5
1.3 The four movements	6
2 The Honest Ledger	7
I Theory	8
3 Preliminaries: the machinery	9
3.1 Representable functors and the Yoneda lemma	9
3.2 Density, the extension functor, and Fam(Setop)	10
3.3 Directed containers and the composition operator $\triangleleft$	11
3.4 Day convolution and the four monoidal structures	12
3.5 Cohomological conventions	13
4 The equivalence chain	14
4.1 One object, four dialects	14
4.2 Containers and the extension	14
4.3 Directed containers	15
4.4 Directed containers are comonads (M2)	15
4.5 The object dictionary: directed containers are small categories (M4)	16
4.6 Polynomial comonoids (M5, bridge)	16
4.7 The chain, assembled; and the variance caveat	17
5 Zappa–Szép products and distributive laws as the composition tool	19
5.1 Composition is a weld	19
5.2 The distributive law: the algebra of the weld	20
5.3 The recognition problem and the free-presheaf lemma	21
5.4 The pairwise criterion	22
5.5 A genuinely pairwise obstruction	23
5.6 The bridge: orchestration composites are Zappa–Szép products	24

6	The correctness criterion $(L) \wedge [\omega] = 0 \in H^2$	26
6.1	The thesis equation	26
6.2	The global condition is a degree-two class	26
6.3	The minimal faithful instance: $[\omega] = \varepsilon$, machine-checked	27
6.4	Two ω -sites: degree-1 descent versus degree-2 Schreier	28
7	The H^2-cluster of obstruction theories	29
7.1	Three homes on one object	29
7.2	The refuted unifier, and the honest umbrella	30
7.3	The crown: the cluster is filtered, not a product	31
8	Change-of-base, $\text{Fam}(\mathcal{C}^{\text{op}})$ generality, and the $\{\text{Id}\}$ classification	32
8.1	The base as a degree of freedom	32
8.2	Positive suite I: m -containers, codensity, polynomiality	33
8.3	Positive suite II: the change-of-base census	34
8.4	Positive suite III: the four monoidal structures	35
8.5	Positive suite IV: when does substitution exist over a base?	36
8.6	The linear wall: Vec is neither full nor faithful	37
8.7	The $\{\text{Id}\}$ collapse	38
9	The set-theoretic frontier: Vec-admissibility and the Gap-S arc	40
9.1	From admissibility to a module	40
9.2	The first count: false under CH	42
9.3	The sharp invariant: $\mu_S(M) = \mathfrak{d}$	42
9.4	The sharpened refutation, and why it matters	43
9.5	The residual, honestly	44
II	Applications	46
10	GA / island-model composition	47
11	Blockchain: re-entrancy as a failed distributive law	48
12	Tax computation (Catala)	49
13	Agent orchestration: the harness is a worker	50
14	Security audits as failed distributive laws	51
15	Supply chains	52
III	Formalisation	53
16	Milestones M1–M6: status	54
17	The sorry-free Lean corpus	55
18	H^2-engines and the free/cofree comonad in Lean	56

19	Change-of-base adjunction and collapse theorems in Lean	57
IV	Impact	58
20	Where compositional correctness has economic consequences	59
21	The AI + Lean pipeline: honesty and trust-grade methodology	60
21.1	The problem this solves	60
21.2	The trust-grade ladder	60
21.3	The registry	61
21.4	The Lean statement self-audit	61
21.5	Forcing functions: honesty made structural	62
21.6	“Correction is the theorem”	62
22	External-publication status	64

How to read this document

This is a document being assembled in stages. It compiles at every stage, and the whole of Part I (Theory) is now written in full, together with the front matter and one Part IV chapter: this chapter, the Overview, and the Honest Ledger; the preliminaries (Chapter 3); the equivalence chain (Chapter 4); Zappa–Szép products and distributive laws (Chapter 5); the correctness criterion (Chapter 6); the H^2 -cluster (Chapter 7); change-of-base and the $\{\text{Id}\}$ classification (Chapter 8); the set-theoretic frontier (Chapter 9); and the AI + Lean pipeline (Chapter 21). These are finished prose meant to be read.

The remaining chapters — the whole of Parts II (Applications) and III (Formalisation), and the two other Part IV chapters — are *scaffolded*: each opens with a short genuine lead-in and then carries a machine-readable assembly directive of the form

```
% ASSEMBLE FROM: <path> [grade] [ASSEMBLE|STUB|GAP]
```

naming the source paper(s) whose bodies will be pulled in during a later assembly session, together with the strongest current trust grade of the material and one of three assembly tags: ASSEMBLE (a written source exists and mainly needs merging), STUB (partial material exists; writing is required), or GAP (no artifact yet; must be drafted or framed as illustrative). A reader can therefore tell at a glance which chapters are load-bearing and which are still promissory.

The trust grades. Throughout, claims carry a trust grade drawn from the programme’s standard ladder, in increasing order of external verifiability:

speculative < computed < proved < peer-reviewed < published < lean-verified.

Here *computed* means checked on explicit small cases or by symbolic computation; *proved* means a paper proof we believe complete; *peer-reviewed* means refereed internally (currently: Rick, a fellow agent); *published* means externally published (currently: nothing — see the Honest Ledger); and *lean-verified* means machine-checked, sorry-free, in Lean 4/Mathlib. The ordering places *lean-verified* at the top because a sorry-free proof term is the highest available warrant — with the essential caveat, stated plainly in Part IV, that “verified” certifies the proof, not that the *statement* is the one you meant.

Chapter 1

Overview: One Thesis

1.1 The organizing equation

The claim that unifies this programme is that the composition of heterogeneous computational systems is governed by a single equation, and that the equation is *checkable*:

$$\boxed{\text{a composite is correct} \iff (\mathbf{L}) \wedge [\omega] = 0 \in H^2}$$

The two conjuncts are independent and have different characters.

(L) — **the local condition.** Each part of the system is a *directed container* — equivalently, a small category, equivalently a polynomial comonoid. Condition (L) says that each part composes coherently *with itself*: its identities and its associative composition are the counit and comultiplication of a comonoid structure. This is the content of the equivalence chain

$$\mathbf{Cont} \simeq \mathbf{DCont} \simeq (\text{polynomial comonoids}) \simeq \mathbf{Cat},$$

which four independent lines of work reach from four directions. (L) is a condition on the pieces, in isolation.

$[\omega] = 0$ — **the global condition.** Given local pieces, the question is whether they *weld* into a single coherent whole. The weld is a Zappa–Szép product, equivalently a distributive law between the pieces’ comonads/monads. Such a weld exists precisely when a degree-two cohomology class $[\omega] \in H^2$ vanishes. Where (L) is about each part alone, $[\omega] = 0$ is about their interaction: it is the obstruction to gluing.

Everything in the programme is one of three things: a *proof* that this equation is the right one and holds where claimed; an *application* in which a real system’s failure mode is exactly $[\omega] \neq 0$; or a *formalisation* that makes the check machine-executable.

1.2 Why this is a foundation and not a slogan

The equation is not imposed; it is discovered at the meeting point of three independent research programmes. Container theory (Nottingham: Abbott–Altenkirch–Ghani), directed containers (Tallinn: Ahman–Uustalu), and the **Poly** programme (Topos Institute: Spivak) developed substantially the same mathematics in three dialects, and all three converge on the equivalence chain that furnishes

(L). Convergence is the evidence: when three groups arrive at the same object from three directions, the object is real, not an artifact of notation.

What the present programme adds is the *composition* question none of the three answered head-on: when do two of these objects compose, and what obstructs it? The answer is the second conjunct. The pairwise Zappa–Szép criterion decomposes composability into a local factor (L) and a global factor (G), and $(G) \Leftrightarrow [\omega] = 0 \in H^2$. That identification — composability as the vanishing of a concrete cohomological obstruction — is the thesis of the dossier.

1.3 The four movements

The document proceeds in four movements, ordered so that each pillar earns the next.

I. Theory — the equation and its domain of validity. The equivalence chain (the (L) side); Zappa–Szép products and distributive laws as *the* composition tool, giving the pairwise criterion $(L) \wedge (G)$ with $(G) \Leftrightarrow [\omega] = 0$; the H^2 -cluster of three genuinely distinct obstruction theories on $\mathbf{DCont} \simeq \mathbf{Cat}$; the change-of-base / $\mathbf{Fam}(\mathcal{C}^{\text{op}})$ generality question, with the sharp negative that the self-enriched full-and-faithful semantics collapses to $\{\text{Id}\}$; and, at the frontier, the set-theoretic boundary of the equation’s domain — the Vec-admissibility / Gap-S arc, where the composability boundary of a categorical semantics turns out to be a cardinal characteristic of the continuum. Theory ends on the live edge, not a solved one.

II. Applications — where $[\omega] \neq 0$ costs money. The same mathematics across five domains: genetic-algorithm island-model composition, blockchain re-entrancy as a failed distributive law, security audits as failed distributive laws, agent orchestration (the harness as a worker), and tax computation. Real systems fail exactly when the weld obstruction is nonzero.

III. Formalisation — the check is executable. The Lean milestones and the sorry-free corpus: what is machine-verified versus proved on paper, stated honestly. This is what distinguishes the grant from hand-waving.

IV. Impact — and a note on honesty. Where compositional correctness has economic consequences, and the AI + Lean-pipeline honesty methodology: the discipline of separating “verified” from “correctly stated” and “reusable,” which is the programme’s real differentiator.

Chapter 2

The Honest Ledger

This chapter is deliberately near the front, not buried in an appendix. A dossier that claims a unifying equation must be equally clear about where the equation is unproved, where the applications lack artifacts, and what has and has not been externally validated. Four items:

1. **The Gap-S / Vec-admissibility wall is OPEN.** Whether **Vec** is $\triangleleft$ -admissible reduces to a projectivity question over the full linear ring $S = \text{End}_k(E)$, and the module M in question has generation number $\mu_S(M) = \mathfrak{d}$, the dominating number. We have proved the conjecture (Conj. 6.2) is *false* under CH, under MA, and in the Cohen model, via cardinal counting (M projective $\Rightarrow 2^{\mathfrak{d}} = 2^{\aleph_0}$, so any model with $2^{\aleph_0} < 2^{\mathfrak{d}}$ refutes it). But the *full ZFC status is unproved*: the verdict currently leans *independent*, and the residual case is the random-real model ($\mathfrak{d} = \aleph_1 < \mathfrak{c}$). We do not yet know whether M is projective there. This is the programme’s deepest open problem, and it is open.
2. **Tax computation has NO formal artifact.** The Catala / tax-computation application (§II.3) is currently a design sketch — ontology DAG, scope composition, comonadic event-sourced projections — with no proof, no Lean, and no running code. It is framed throughout as illustrative-only.
3. **Equivalence-chain Lean milestones M5–M6 are unproved.** Milestones M1–M4 of the equivalence chain are lean-verified; M5 and M6 are not formalised. The chain is complete on paper but not machine-checked end to end.
4. **Nothing is externally published.** Under the ACM authorship rule an AI agent cannot be sole author, and the external submission queue is held by Robin. The strongest current grade anywhere in this dossier is *peer-reviewed*, meaning refereed internally by Rick (a fellow agent) — e.g. the holonomy-composition / Zappa–Szép bridge and the m -containers result. No claim here is *published* in the external sense.

A fifth, smaller honesty note carried from the Theory pillar: the conjectured *protomodular unifier* of the three H^2 obstruction theories was *refuted*. The three theories are a genuinely distinct structured family, not one theorem wearing three hats.

Part I

Theory

Chapter 3

Preliminaries: the machinery

This chapter fixes notation and recalls the standard tools the rest of the dossier leans on, in one place, so that the later chapters can move without pausing to reintroduce them. *Almost everything here is standard* and is collected for self-containedness: the Yoneda lemma is Yoneda’s and Mac Lane’s; density and the coend/Kan calculus follow Mac Lane [3], Riehl [4], and Loregian [2]; Day convolution is Day’s [1]. The only programme-internal content is the *container reading* of each tool — the translation into shapes and positions — which is *proved* and is what makes the later theorems computable. Two facts of Ghani’s are foregrounded because the composition machinery uses them as black boxes: that the container-extension functor *is* a left Kan extension along the representable embedding (§3.2), and that the closed structure of a Day category is *forced on representables and extended by density* (§3.4).

The reader who wants a map: §3.1–3.2 (representables, Yoneda, density, the extension functor) are used everywhere the (L) side appears — the equivalence chain of Chapter 4, and, in an essential and less obvious way, the module reduction of the set-theoretic frontier (Chapter 9), where $\text{Nat}(F, -) \cong \text{Hom}_S(F(E), -)$ is Yoneda in disguise. Section 3.3 fixes the directed-container and composition ($\triangleleft$) notation carried by the Zappa–Szép chapters. Section 3.4 (Day convolution) supports the change-of-base chapter (Chapter 8). Section 3.5 fixes the cohomological conventions — H^2 , the class $[\omega]$, the Lyndon–Hochschild–Serre transgression — on which the correctness criterion and the H^2 -cluster of Chapter 7 stand.

Conventions. **Set** is the category of sets and functions. For a locally small category $\mathcal{C}$ we write $\mathcal{C}(a, b)$ for its hom-set; $\text{ob } \mathcal{C}$ for its objects. Functors are covariant unless a superscript op says otherwise. A *presheaf* on $\mathcal{C}$ is a functor $\text{Cop} \rightarrow \mathbf{Set}$; a *copresheaf* is a functor $\mathcal{C} \rightarrow \mathbf{Set}$. We write $\cong$ for canonical (natural) isomorphism and are careful to say *which* isomorphism when it matters.

3.1 Representable functors and the Yoneda lemma

Every construction in the dossier eventually reduces to a hom-set. The representable functors are the functors that *are* a hom-set with one argument held fixed, and they generate everything else.

Definition 3.1 (Representable copresheaf). Fix a set A . The *covariant representable* at A is

$$y^A := \mathbf{Set}(A, -) : \mathbf{Set} \longrightarrow \mathbf{Set}, \quad y^A(X) = \mathbf{Set}(A, X) = X^A,$$

sending $f : X \rightarrow Y$ to post-composition $y^A(f) = f \circ (-)$. A copresheaf $F : \mathbf{Set} \rightarrow \mathbf{Set}$ is *representable* if $F \cong y^A$ for some A , the *representing object*.

Remark 3.2 (The container reading). A *container* is a pair (S, P) of a set S of *shapes* and a family $P: S \rightarrow \mathbf{Set}$ of *position sets*, with *extension*

$$\llbracket S, P \rrbracket : \mathbf{Set} \rightarrow \mathbf{Set}, \quad \llbracket S, P \rrbracket(X) = \sum_{s \in S} \mathbf{Set}(P(s), X).$$

The *representable container* has a single shape and A positions, $(S, P) = (1, A)$, and $\llbracket 1, A \rrbracket(X) = \mathbf{Set}(A, X) = y^A(X)$. So representable copresheaves are exactly the extensions of single-shape containers, and the coproduct in the general formula is a coproduct of *representables* indexed by shapes. This is the whole story, phrased below first as a Kan extension (§3.2) and then as a Day convolution (§3.4). The same extension formula, over a base $\mathcal{C}$ other than $\mathbf{Set}$, is the object of study of Chapter 9.

Definition 3.1 is uniform in A : a function $a: A' \rightarrow A$ induces, by *pre-composition*, a natural transformation $y^A \Rightarrow y^{A'}$. Thus $A \mapsto y^A$ is a functor $y^{(-)}: \mathbf{Setop} \rightarrow [\mathbf{Set}, \mathbf{Set}]$, the *Yoneda embedding*. It is genuinely an embedding, as the Yoneda lemma shows.

Theorem 3.3 (Yoneda lemma; standard). *For any functor $F: \mathbf{Set} \rightarrow \mathbf{Set}$ and any set A there is a bijection*

$$\Theta_{F,A} : \mathbf{Nat}(y^A, F) \xrightarrow{\cong} F(A), \quad \Theta(\alpha) = \alpha_A(\mathrm{id}_A),$$

natural in F and A .

Proof. The inverse sends $x \in F(A)$ to $\Psi(x)$ with $\Psi(x)_X(g) = F(g)(x)$ for $g: A \rightarrow X$; naturality of $\Psi(x)$ is functoriality, $F(h)(F(g)(x)) = F(h \circ g)(x)$. Then $\Theta(\Psi(x)) = F(\mathrm{id}_A)(x) = x$. Conversely, for $\alpha: y^A \Rightarrow F$ and $x = \alpha_A(\mathrm{id}_A)$, naturality of α at $g: A \rightarrow X$ chased on id_A gives $\alpha_X(g) = F(g)(x) = \Psi(x)_X(g)$, so $\Psi(x) = \alpha$. Naturality in F and A is the same chase. $\square$

Corollary 3.4 (Full faithfulness). $\mathbf{Nat}(y^A, y^B) \cong \mathbf{Set}(B, A)$, *naturally; the Yoneda embedding $y^{(-)}: \mathbf{Setop} \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is fully faithful. Natural transformations between representables are exactly functions, in the opposite direction.*

Corollary 3.4 is the reason the container-morphism convention (Notation 3.6) has position maps running *backwards*: a map of representables is a backwards map of representing objects.

3.2 Density, the extension functor, and $\mathbf{Fam}(\mathbf{Setop})$

The Yoneda lemma says representables see everything. Density says they *are* everything, once colimits are allowed.

Theorem 3.5 (Density / co-Yoneda; standard). *Every functor $F: \mathbf{Set} \rightarrow \mathbf{Set}$ is canonically a colimit of representables:*

$$F \cong \int^{A \in \mathbf{Set}} F(A) \bullet y^A,$$

where $F(A) \bullet y^A$ is the $F(A)$ -fold copower of y^A . Equivalently F is the colimit of the representables $y^A \xrightarrow{\Psi(x)} F$ over the category of elements (A, x) , $x \in F(A)$.

Reference. The density formula, dual to Yoneda; Riehl [4, §6.2], Loregian [2, Ch. 2]. Its universal property is Theorem 3.3 read backwards: a cocone from $F(A) \bullet y^A$ to G is a family $F(A) \rightarrow \mathbf{Nat}(y^A, G) \cong G(A)$ natural in A , i.e. a natural transformation $F \Rightarrow G$. $\square$

The representables are thus a *dense generator* of $[\mathbf{Set}, \mathbf{Set}]$: two constructions that agree on representables and preserve the relevant colimits agree everywhere. This uniformity is the workhorse of the whole (L) side.

Notation 3.6 (**Cont** and $\mathbf{Fam}(\mathbf{Setop})$). A morphism of containers $(S, P) \rightarrow (T, Q)$ is a pair (u, v) with $u: S \rightarrow T$ a map of shapes and v a family $v_s: Q(us) \rightarrow P(s)$ of position maps *going backwards*. This is exactly an S -indexed family of objects of $\mathbf{Setop}$ together with a map of index sets and, over it, morphisms in $\mathbf{Setop}$; hence

$$\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Setop}),$$

the *free coproduct completion* of $\mathbf{Setop}$ — the universal category with small coproducts containing $\mathbf{Setop}$. Under $\llbracket - \rrbracket$, a family $(P(s))_{s \in S}$ maps to $\sum_{s \in S} y^{P(s)}$.

Theorem 3.7 (Extension is a left Kan extension; Ghani’s observation, proved). *Let $\mathcal{A}$ be a category and $Y: \mathcal{A} \rightarrow \mathbf{Fam}(\mathcal{A})$ the singleton embedding $a \mapsto (a)$. For any $F: \mathcal{A} \rightarrow \mathcal{E}$ into a category with small coproducts, the coproduct-preserving extension $\widehat{F}((a_s)_{s \in S}) = \coprod_{s \in S} F(a_s)$ is the left Kan extension $\mathbf{Lan}_Y F$, and it preserves representables ($\widehat{F} \circ Y \cong F$). Specialising $\mathcal{A} = \mathbf{Setop}$, $F = y^{(-)}$, and $\mathcal{E} = [\mathbf{Set}, \mathbf{Set}]$, the container-extension functor is that Kan extension:*

$$\llbracket - \rrbracket = \mathbf{Lan}_Y y^{(-)}, \quad \llbracket S, P \rrbracket = \coprod_{s \in S} y^{P(s)}.$$

Proof. By the pointwise coend formula [3, Ch. X] and the description of morphisms out of a singleton in a free coproduct completion, $\mathbf{Fam}(\mathcal{A})(Ya, (a_s)_s) \cong \coprod_s \mathcal{A}(a, a_s)$, so

$$(\mathbf{Lan}_Y F)((a_s)_s) \cong \int^a \left(\coprod_s \mathcal{A}(a, a_s) \right) \bullet F(a) \cong \coprod_s \int^a \mathcal{A}(a, a_s) \bullet F(a) \cong \coprod_s F(a_s),$$

the inner coend being density (Theorem 3.5) at a_s . Preservation of representables is the one-term case $S = 1$. For $\mathcal{A} = \mathbf{Setop}$ and $F = y^{(-)}$ this reads $\llbracket S, P \rrbracket = \coprod_s y^{P(s)}$, whose value at X is $\sum_s \mathbf{Set}(P(s), X) = \llbracket S, P \rrbracket(X)$ (Remark 3.2). A size remark: although $\mathbf{Setop}$ is not small, part of the statement is really the universal property of the free coproduct completion, which holds verbatim, and the coproducts landed on are small since shape sets are. $\square$

Two consequences are used without reproof throughout Part I: (i) any structure on $\mathbf{Setop}$ transported to **Cont** need only be defined on singletons and extended by coproducts, and the result is automatically the Kan extension; (ii) to check a functor out of **Cont** agrees with $\llbracket - \rrbracket$ it suffices to check agreement on representables plus preservation of coproducts. “A container is a coproduct of representables” is not a slogan but a computational method.

3.3 Directed containers and the composition operator $\triangleleft$

The extension functors above are the raw material; the (L) side of the correctness criterion asks which of them compose *with themselves* coherently. That is the content of directed containers, whose full theory — including the equivalence chain — is Chapter 4. Here we fix only the notation the composition chapters need.

The category **Cont** carries a monoidal structure $\triangleleft$, *composition*, that on extensions is composition of endofunctors: $\llbracket p \triangleleft q \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$. Concretely, with $p = (S, P)$ and $q = (T, Q)$, a shape of $p \triangleleft q$ is a shape $s \in S$ together with a $P(s)$ -indexed family of shapes of q , and a position is a dependent pair — the “a shape, then a shape at each position” law that makes $\triangleleft$ the substitution tensor. The unit is the identity container $y^1 = (1, 1)$, whose extension is the identity functor.

Definition 3.8 (Directed container, informally). A *directed container* is a container p equipped with the structure of a $\triangleleft$ -comonoid: a counit $p \rightarrow y^1$ and a comultiplication $p \rightarrow p \triangleleft p$ satisfying the comonoid laws. Equivalently — and this is the equivalence chain — it is a small category, a polynomial comonoid in **Poly**, or a cofunctor domain, one object presented four ways. We write **DCont** for the category of directed containers.

Theorem 3.9 (The equivalence chain; assembled in Chapter 4). *There are equivalences of categories*

$$\mathbf{Cont} \supseteq \mathbf{DCont} \simeq (\text{polynomial comonoids in } \mathbf{Poly}) \simeq \mathbf{Cat},$$

where **Cat** is the category of small categories and (co)functors. The directed-container comonoid data are exactly a category's identities and composition.

This is the (L) condition: a system is composable-with-itself precisely when it is a directed container, equivalently a small category. The proof, and the four independent lines of work that converge on it, are the subject of Chapter 4; we state it here only so that later chapters may say “directed container” and “small category” interchangeably.

3.4 Day convolution and the four monoidal structures

Besides $\triangleleft$, the category **Cont** carries tensors transported from monoidal structures on the base by *Day convolution*. These are the setting of the change-of-base chapter (Chapter 8); we record only the degeneration that makes them coend-free.

Definition 3.10 (Day convolution; standard). For a small monoidal category $(\mathcal{C}, \otimes, I)$ the *Day convolution* of copresheaves F, G is

$$(F \otimes G)(c) = \int^{c_1, c_2} \mathcal{C}(c_1 \otimes c_2, c) \times F(c_1) \times G(c_2),$$

with unit y^I . Then $([\mathcal{C}, \mathbf{Set}], \otimes, y^I)$ is monoidal, cocontinuous in each variable, and $y^{(-)}$ is strong monoidal: $y^a \otimes y^b \cong y^{a \otimes b}$ [1].

Proposition 3.11 (Degenerate Day on containers; proved). *For any monoidal structure $\star$ on **Set**, Day convolution induces on $\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Setop})$ a monoidal structure $\odot_\star$ given without any coend by*

$$p \odot_\star q = (S_p \times S_q, (s, t) \mapsto p[s] \star q[t]),$$

where $p = (S_p, p[-])$, $q = (S_q, q[-])$. Shapes multiply; only the position law changes.

Proof. Write $p = \coprod_s y^{p[s]}$, $q = \coprod_t y^{q[t]}$ (Theorem 3.7). Day is cocontinuous in each variable, so $p \odot_\star q \cong \coprod_{s,t} (y^{p[s]} \odot_\star y^{q[t]}) \cong \coprod_{s,t} y^{p[s] \star q[t]}$ by strong monoidality on representables; read off the container by Theorem 3.7. The coend has been discharged on the generators. $\square$

Taking $\star \in \{+, \times\}$ gives two of the four tensors on **Cont** (the additive and Dirichlet convolutions); the substitution tensor $\triangleleft$ of §3.3 is a third, and it is *not* a Day convolution. The classification — which tensors arise, and which are closed — is Chapter 8. One fact from there is used as a black box: each Day tensor is closed, and its internal hom is *forced* on representables, $[y^a, H]_\otimes \cong H((-) \otimes a)$, then extended by density; for the Dirichlet tensor $[y^a, y^b]$ is the “shapes are morphisms” container with shape set $\mathbf{Cont}(y^a, y^b)$. Density *determines* this internal hom but does not hand it to every cocontinuous functor for free — preservation of a right adjoint is a separate, checkable fact — which is exactly the subtlety the $\{\text{Id}\}$ collapse of Chapter 8 turns on.

3.5 Cohomological conventions

The global condition $[\omega] = 0$ is a statement in low-degree cohomology. We fix conventions here; the theorems are Chapters 5–7.

For a group G and a G -module M we write $H^2(G; M)$ for the second group cohomology, classifying extensions $1 \rightarrow M \rightarrow E \rightarrow G \rightarrow 1$ up to equivalence; the class of an extension is its *Schreier class*. For a groupoid $\mathcal{G}$ we use the cohomology of its classifying space, which for a *connected* groupoid reduces to that of its vertex group π : $H^2(\mathcal{G}; M) \cong H^2(\pi; M)$. The class we call $[\omega]$ is the obstruction to a coherent weld of two systems — the Schreier class of the Zappa–Szép extension realising their composite; the criterion $(G) \iff [\omega] = 0$ is Chapter 6.

Two structural tools recur. For a normal extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ the *Lyndon–Hochschild–Serre spectral sequence* has $E_2^{p,q} = H^p(\Gamma; H^q(\mathcal{D}; M))$ converging to $H^{p+q}(\pi; M)$; its degree-two *transgression* d_2 is the differential that, in the programme’s identification, carries the holonomy ($d_2(\varphi) = \varphi_*[\omega]$). And the phrase “ $[\omega]$ is a differential, not a summand” means precisely that a base-edge class fails to survive to the abutment when $d_2 \neq 0$. These are the load-bearing facts behind the “filtration, not product” grammar of the H^2 -cluster (Chapter 7); we invoke them there by name.

Chapter 4

The equivalence chain

The local condition (L) rests on the identification of four objects that were developed independently: ordinary containers, directed containers, polynomial comonoids, and small categories. This chapter assembles the chain $\mathbf{Cont} \simeq \mathbf{DCont} \simeq (\text{poly comonoids}) \simeq \mathbf{Cat}$ and fixes the notation used throughout the dossier. It is the foundation on which every later composition statement stands: to compose systems coherently, each system must first *be* one of these objects.

4.1 One object, four dialects

The organising slogan of the chapter is that a small category, a directed container, and a polynomial comonoid are *the same object spoken in three dialects*, sitting inside the larger world of containers as exactly those that carry comonad structure. The four vocabularies line up term for term.

Notation 4.1 (The four dialects). Throughout, we translate freely across the following dictionary.

Container	Directed container	Polynomial	Category
shape $s \in S$	shape s	position $s \in p(1)$	object s
position $p \in P(s)$	position p	direction $p \in p[s]$	morphism out of s
—	root $\mathbf{o}(s)$	—	identity id_s
—	sub-shape $s \downarrow p$	—	codomain $\text{cod}(p)$
—	shift $p \oplus q$	—	composite $q \circ p$
container map	directed-container map	comonoid hom.	<i>cofunctor</i>

The bottom row is the one subtlety carried by this chapter: the equivalences of the chain are equivalences *of categories* between directed containers and polynomial comonoids, whereas the identification with $\mathbf{Cat}$ is a bijection *on objects*; the native morphisms are cofunctors, not functors (§4.7).

The plan is: recall containers and their extension (§4.2); add the directed structure (§4.3); show that directed structure is *exactly* comonad structure on the extension (§4.4); read that comonad as a small category (§4.5); re-read the same data in **Poly** as a comonoid (§4.6); and assemble the chain with its variance caveat and formalisation status (§4.7).

4.2 Containers and the extension

We recall only what the chain uses; the definitions and their basic theory are established in Chapter 3. A *container* is a pair (S, P) of a set S of *shapes* and a family $P: S \rightarrow \mathbf{Set}$ of *positions* [24].

Its *extension* is the endofunctor

$$\llbracket S, P \rrbracket(X) = \sum_{s \in S} X^{P(s)} = \sum_{s \in S} \mathbf{Set}(P(s), X),$$

an element being a pair (s, v) with $v: P(s) \rightarrow X$. As recorded in the preliminaries, containers organise into $\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Set}^{\text{op}})$ (Notation 3.6), a morphism $(S, P) \rightarrow (S', P')$ being a shape map $f: S \rightarrow S'$ *forward* together with a position map $f_s^\sharp: P'(fs) \rightarrow P(s)$ *backward*; the extension is the left Kan extension exhibited in Theorem 3.7, and $\llbracket - \rrbracket: \mathbf{Cont} \rightarrow \mathbf{End}(\mathbf{Set})$ is full and faithful onto the polynomial functors by the Yoneda/density argument of Theorems 3.3–3.5. We do not reprove these here.

Remark 4.2 (The backward variance is the whole story). The contravariance of $f^\sharp$ in positions is not incidental. It is the fibrewise opposite of $\mathbf{Fam}(\mathbf{Set}^{\text{op}})$, and it is precisely what will force the morphisms of directed containers to be cofunctors rather than functors (§4.7). Every phenomenon in this chapter is a consequence of positions running backward while shapes run forward.

4.3 Directed containers

Chapter 3 introduced a directed container informally as a $\triangleleft$ -comonoid (Definition 3.8). We now fix the explicit operational form used for all computations, following Ahman–Chapman–Uustalu [25].

Definition 4.3 (Directed container). A *directed container* is a container (S, P) equipped with

- a *root* $\mathbf{o}(s) \in P(s)$ for each $s \in S$;
- a *sub-shape* operation $s \downarrow p \in S$ for $p \in P(s)$;
- a *shift* $p \oplus q \in P(s)$ for $p \in P(s)$ and $q \in P(s \downarrow p)$;

subject, for all $s, p \in P(s)$, $q \in P(s \downarrow p)$ and $q' \in P((s \downarrow p) \downarrow q)$, to

$$\begin{aligned} \text{(D1)} \quad s \downarrow \mathbf{o}(s) &= s, & \text{(D2)} \quad \mathbf{o}(s) \oplus q &= q, & \text{(D3)} \quad p \oplus \mathbf{o}(s \downarrow p) &= p, \\ \text{(D4)} \quad s \downarrow (p \oplus q) &= (s \downarrow p) \downarrow q, & \text{(D5)} \quad (p \oplus q) \oplus q' &= p \oplus (q \oplus q'). \end{aligned}$$

Directed containers and their morphisms (Definition 4.14) form the category $\mathbf{DCont}$.

Remark 4.4 (An indexed monoid). Laws (D2), (D3), (D5) make each $P(s)$ a monoid-like structure with unit $\mathbf{o}(s)$; (D1) and (D4) track how sub-shapes nest. The typing of (D5) depends on (D4): for $p \oplus (q \oplus q')$ to be defined one needs $q \oplus q' \in P(s \downarrow p)$ with $(s \downarrow p) \downarrow (q \oplus q') = s \downarrow (p \oplus q)$, which is exactly (D4). This foreshadows the categorical reading, where the well-typedness of associative composition rests on the codomain law.

4.4 Directed containers are comonads (M2)

The first link identifies directed structure with comonad structure on the extension. This is the content the roadmap calls milestone M2.

Proposition 4.5 (The extension comonad, computed; *proved, lean-verified*). *Let (S, P) be a directed container and $D := \llbracket S, P \rrbracket$. Define*

$$\varepsilon_X(s, v) = v(\mathbf{o}(s)), \quad \delta_X(s, v) = (s, \lambda p. (s \downarrow p, \lambda q. v(p \oplus q))).$$

Then (D, ε, δ) is a comonad, and each comonad law is exactly one directed-container law: the left counit is (D1)+(D2), the right counit is (D3), and coassociativity is (D4)+(D5). Conversely, every comonad structure on $\llbracket S, P \rrbracket$ arises this way, so directed-container structures on (S, P) correspond bijectively to comonad structures on $\llbracket S, P \rrbracket$.

Proof sketch. Write $\delta(s, v) = (s, w)$ with $w(p) = (s \downarrow p, \lambda q. v(p \oplus q))$. *Left counit:* $\varepsilon(s, w) = w(\mathbf{o}(s)) = (s \downarrow \mathbf{o}(s), \lambda q. v(\mathbf{o}(s) \oplus q)) = (s, v)$ using (D1) for the shape and (D2) for the value. *Right counit:* $\varepsilon(w(p)) = v(p \oplus \mathbf{o}(s \downarrow p)) = v(p)$ by (D3). *Coassociativity:* both composites of δ reduce to $(s, \lambda p. (s \downarrow p, \lambda q. ((s \downarrow p) \downarrow q, \lambda q'. v(p \oplus (q \oplus q')))))$, the inner shape agreeing by (D4) and the value by (D5). The converse is the reverse reading: full-and-faithfulness of $\llbracket - \rrbracket$ (Theorem 3.3) makes the displayed computations force (D1)–(D5) from the comonad laws. $\square$

Remark 4.6 (Provenance and formalisation). The biconditional is Ahman–Chapman–Uustalu [25, Thm. 3.15]: the category of directed containers is the pullback of the forgetful functor $\mathbf{Comon}(\mathbf{EndSet}) \rightarrow \mathbf{End}(\mathbf{Set})$ along $\llbracket - \rrbracket$. The coordinate proof of the comonad laws above is machine-checked in Lean (the position-transfer comonad-law calculation, `[Quot.sound]`-only) [51]. We report this as *lean-verified* at the level of the law-matching; the full pullback statement of [25] is *proved* in the source and not independently re-formalised here.

4.5 The object dictionary: directed containers are small categories (M4)

The second link reads the comonad of §4.4 as a small category. This is Ahman–Uustalu [26]; we record the bijection of data.

Theorem 4.7 (Object dictionary [26]; *proved*). *Directed-container structures on (S, P) are in bijection with small categories $\mathcal{C}$ with object set S and $P(s) = \coprod_{s'} \mathcal{C}(s, s')$ (the morphisms out of s), under*

$$\text{ob}\mathcal{C} = S, \quad \mathcal{C}(s, s') = \{p \in P(s) : s \downarrow p = s'\}, \quad \text{id}_s = \mathbf{o}(s), \quad q \circ p = p \oplus q.$$

In particular $s \downarrow p = \text{cod}(p)$, and every small category arises from a unique directed container.

Proof sketch. (Category $\Rightarrow$ directed container.) Set $P(s) = \coprod_{s'} \mathcal{C}(s, s')$, $s \downarrow p = \text{cod}(p)$, $\mathbf{o}(s) = \text{id}_s$, $p \oplus q = q \circ p$. Then (D1) is $\text{cod}(\text{id}_s) = s$; (D2), (D3) are the unit laws; (D4) is $\text{cod}(q \circ p) = \text{cod}(q)$; (D5) is associativity, written with the order reversal $p \oplus q = q \circ p$. (Directed container $\Rightarrow$ category.) Take $\mathcal{C}(s, s') = \{p \in P(s) : s \downarrow p = s'\}$, identities $\mathbf{o}(s)$ (landing in $\mathcal{C}(s, s)$ by (D1)), and composite $p \oplus q$ (landing in $\mathcal{C}(s, s'')$ by (D4)); unit laws are (D2), (D3), associativity is (D5). The two constructions are mutually inverse. $\square$

Corollary 4.8 (The two atoms). *One-shape directed containers ($|S| = 1$) are exactly monoids; thin ones ($|\mathcal{C}(s, s')| \leq 1$) are exactly preorders. Every small category is assembled from these two phenomena — many objects and endo-loops.*

Example 4.9 (The two generating cases). For the poset $\mathbf{2} = \{0 \xrightarrow{u} 1\}$: $S = \{0, 1\}$, $P(0) = \{\mathbf{o}(0), u\}$, $P(1) = \{\mathbf{o}(1)\}$, with $0 \downarrow u = 1$ and $u \oplus \mathbf{o}(1) = u$; the extension has $|\llbracket S, P \rrbracket(1)| = 3$, the three morphisms. For the monoid $\mathbb{Z}/3$: one shape $\star$, $P(\star) = \mathbb{Z}/3$, $\star \downarrow p = \star$, $\mathbf{o}(\star) = 0$, $p \oplus q = p + q \bmod 3$; the extension is the shift comonad $DX = X^3$ with $\varepsilon(v) = v(0)$ and $\delta(v) = \lambda p. \lambda q. v(p + q)$. All laws are verified by hand in the source.

4.6 Polynomial comonoids (M5, bridge)

The third dialect is Spivak’s category **Poly** of polynomial endofunctors on **Set**. A container (S, P) is the polynomial $p = \sum_{s \in S} y^{P(s)}$ with $p(1) = S$ and $p[s] = P(s)$; container maps are maps of polynomials, so **Cont** is **Poly** under a change of name [31, 32]. The composition tensor $\triangleleft$ on **Poly** corresponds to composition of extensions, $\llbracket p \triangleleft q \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$.

Theorem 4.10 (Polynomial comonoids are categories [29, 30, 31]; *proved* in the sources). *The comonoids for $(\mathbf{Poly}, \triangleleft, y^{(-)})$ — polynomial comonoids — are exactly small categories: the counit is the identity-assigning map and the comultiplication is composition, subject to the comonoid laws which are the category axioms. Their morphisms are the comonoid homomorphisms, and the resulting category is $\mathbf{Cat}^\sharp = \mathbf{Comod}(\mathbf{Poly})$.*

Remark 4.11 (One structure, two derivations). A polynomial comonoid on $p = \llbracket S, P \rrbracket$ is the same data as the extension comonad of Proposition 4.5: comultiplication δ is $s \mapsto (s \downarrow p, \oplus)$ and counit ε is $\mathbf{o}$. Ahman–Uustalu [25, 26] and Spivak et al. [29, 30] arrived at this identification independently, from datatypes and from interaction respectively. The translation is tabulated in Notation 4.1. This is milestone M5; the full agreement of *all* monoidal structures on **Cont** with the four products of **Poly** (coproduct, product, $\triangleleft$, Dirichlet) is *open* and not formalised.

4.7 The chain, assembled; and the variance caveat

Combining the links gives the chain of the chapter, which we now state precisely — and correct where the informal slogan misleads.

Theorem 4.12 (The equivalence chain; *proved*). *There are equivalences of categories*

$$\mathbf{DCont} \simeq \mathbf{Comod}(\mathbf{Poly}) = \mathbf{Cat}^\sharp,$$

and a bijection, natural in the data, between the objects of these categories and small categories (Theorem 4.7). Directed containers embed into containers by the forgetful functor $\mathbf{DCont} \rightarrow \mathbf{Cont}$ that discards $(\mathbf{o}, \downarrow, \oplus)$; this functor is faithful and injective on objects but not full, so $\mathbf{DCont}$ is a proper (non-full) subclass of $\mathbf{Cont}$ — exactly the containers whose extension carries a comonad structure (Proposition 4.5).

Remark 4.13 (Reading the chain of the chapter head). The displayed chain $\mathbf{Cont} \simeq \mathbf{DCont} \simeq (\text{poly comonoids}) \simeq \mathbf{Cat}$ must be read with two qualifications, both proved above.

1. The link **Cont–DCont** is a *containment* $\mathbf{Cont} \supseteq \mathbf{DCont}$ (the forgetful functor of Theorem 4.12), not an equivalence: not every container is directed.
2. The link to **Cat** is an equivalence *on objects* only. As a category, $\mathbf{DCont} \simeq \mathbf{Cat}^\sharp$, whose morphisms are cofunctors, not functors (Theorem 4.16 below).

The genuine equivalences of categories are $\mathbf{DCont} \simeq \mathbf{Comod}(\mathbf{Poly}) = \mathbf{Cat}^\sharp$.

We make the second qualification precise, since it governs how systems compose in every later chapter. The maps native to directed containers pull positions backward, $f_s^\sharp: P'(fs) \rightarrow P(s)$ (Remark 4.2), which under the dictionary lifts a morphism out of fs to one out of s — the variance of a *cofunctor*, not a functor.

Definition 4.14 (Directed-container morphism). A morphism $D \rightarrow D'$ of directed containers is a container morphism $(f, f^\sharp)$ satisfying, for all $s, h \in P'(fs)$ and $k \in P'((fs) \downarrow' h)$,

$$\begin{aligned} \text{(M0)} \quad & f(s \downarrow f_s^\sharp(h)) = (fs) \downarrow' h, \\ \text{(M1)} \quad & f_s^\sharp(\mathbf{o}'(fs)) = \mathbf{o}(s), \\ \text{(M2)} \quad & f_s^\sharp(h \oplus' k) = f_s^\sharp(h) \oplus f_{s \downarrow f_s^\sharp(h)}^\sharp(k). \end{aligned}$$

Definition 4.15 (Cofunctor [34, 35, 29]). A *cofunctor* $\varphi: \mathcal{C} \bullet \mathcal{D}$ is an object map φ_0 together with a lift assigning to each object c and morphism $h: \varphi_0 c \rightarrow d$ a morphism $\varphi^\uparrow(c, h): c \rightarrow c'$, subject to

(C0) $\varphi_0(\text{cod } \varphi^\uparrow(c, h)) = \text{cod}(h)$, (C1) $\varphi^\uparrow(c, \text{id}_{\varphi_0 c}) = \text{id}_c$, and (C2) $\varphi^\uparrow(c, h' \circ h) = \varphi^\uparrow(c', h') \circ \varphi^\uparrow(c, h)$ with $c' = \text{cod } \varphi^\uparrow(c, h)$. Small categories and cofunctors form $\mathbf{Cof} = \mathbf{Cat}^\sharp$, the comonoids in $(\mathbf{Poly}, \triangleleft)$ and their homomorphisms [29, 30].

Theorem 4.16 (The morphism dictionary [26, 30]; *proved, lean-verified*). *The assignment $(f, f^\sharp) \mapsto (\varphi_0 := f, \varphi^\uparrow(s, h) := f_s^\sharp(h))$ is an isomorphism of categories*

$$\mathbf{DCont} \cong \mathbf{Cof},$$

*matching (M0)–(M2) to (C0)–(C2) term for term under the substitutions “position out of $s =$ morphism out of s ” and “ $x \oplus y = y \circ x$ ”. This isomorphism is formalised in Lean 4 with zero **sorry** and zero axioms [51].*

Proof sketch. Under the object dictionary the two sides are literally the same data: f is an object map and $f^\sharp$ is a lift. (M0) is the codomain law (C0); (M1) is the unit law (C1); and, writing $p = f_s^\sharp(h)$, $s' = s \downarrow p$, the shift law (M2) reads $f_s^\sharp(k \circ' h) = \varphi^\uparrow(s', k) \circ \varphi^\uparrow(s, h)$, which is (C2). The correspondence is a bijection on every hom-set and preserves identities and composition (position maps compose backward, $(g \circ f)_s^\sharp = f_s^\sharp \circ g_{f_s}^\sharp$), and it is a bijection on objects by Theorem 4.7; a bijective-on-objects fully faithful functor is an isomorphism of categories. $\square$

Proposition 4.17 (The dictionary does not extend to $\mathbf{Cat}$ with functors; *proved* [26, 30]). *There is no functor $\mathbf{DCont} \rightarrow \mathbf{Cat}$ agreeing with the object dictionary on objects and full. Already for the terminal category $\mathbf{1}$ and $\mathbf{2} = \{0 \rightarrow 1\}$ there is exactly one cofunctor $\mathbf{1} \bullet \mathbf{2}$ but two functors $\mathbf{1} \rightarrow \mathbf{2}$, so $|\mathbf{DCont}(D_1, D_2)| = 1 < 2 = |\mathbf{Cat}(\mathbf{1}, \mathbf{2})|$. The comparison is not even monotone: $|\mathbf{Cat}(\mathbf{3}, \mathbf{2})| = 4 < 5 = |\mathbf{Cof}(\mathbf{3}, \mathbf{2})|$ while $|\mathbf{Cat}(\mathbf{2}, \mathbf{2})| = 3 > 2 = |\mathbf{Cof}(\mathbf{2}, \mathbf{2})|$, so neither hom-set contains the other.*

Remark 4.18 (Why the variance is the useful half). The covariant variant (position maps $P(s) \rightarrow P'(fs)$ under the mirror-image conditions) recovers functors exactly, but these are *not* morphisms of the underlying container. The contravariant, native half is the *Put* of a delta lens [35]: a directed-container morphism is the update-propagating component of a bidirectional transformation. Every later composition statement in this dossier — supply-chain stages, agent hierarchies, state updates — composes its maps as cofunctors; reading them as functors keeps the *Get* and silently discards the *Put*, and even changes the count of maps (Proposition 4.17).

Remark 4.19 (Formalisation status of the chain). Against the roadmap milestones, the object-level chain is verified as follows. *M1* (containers, the representation theorem) is standard and recorded in Chapter 3. *M2* (directed container = comonad, Proposition 4.5) is *proved* in [25] with the law-matching *lean-verified* [51]. *M4* ($\mathbf{DCont} \cong \mathbf{Cof}$, Theorem 4.16) is *lean-verified*, zero **sorry** and zero axioms [51]. Milestones *M5* (agreement of all four monoidal products with **Poly**, Remark 4.11) and *M6* (the double category $\mathbf{Org}$ of orchestration) are *open*: stated in the sources as targets, not yet formalised, and not claimed here.

Chapter 5

Zappa–Szép products and distributive laws as the composition tool

Given local pieces, composition is a *weld*, and the canonical weld of two categories/comonads is the Zappa–Szép product, equivalently a distributive law. This chapter develops the pairwise criterion: two directed containers compose iff the local condition (L) and a global condition (G) both hold. It is the machinery that makes “ $[\omega] = 0$ ” precise in the next chapter.

5.1 Composition is a weld

Fix the standing dictionary of the preliminaries: a directed container is a $\triangleleft$ -comonoid, and **DCont** $\cong$ **Cat** identifies it with a small category (Definition 3.8, the chain of Theorem 3.9). Under this identification the objects of the container are its shapes, the morphisms out of a shape are its positions, identities are roots, and composition is position-concatenation [26].

The question of this chapter is not how to run *one* directed container — a functor already does that, and a functor cannot fail — but how to *combine two*. Running two compositional systems that share a resource is the *interleaving* of two handoff categories, and interleaving is not composition along a fixed topology. It is a weld: the two systems must be fitted together inside a single third system so that each morphism of the whole decomposes, uniquely and coherently, into a piece from one factor followed by a piece from the other. The canonical such weld is the Zappa–Szép product, and its structural carrier is a distributive law.

Definition 5.1 (Wide subcategory, strict factorisation). Throughout $\mathcal{K}$ is a small category and every subcategory is *wide*: $\mathcal{C} \subseteq \mathcal{K}$ is wide if $\text{ob}\mathcal{C} = \text{ob}\mathcal{K}$, it is closed under composition, and it contains every identity. Write dom , cod for source and target. Given wide $\mathcal{C}, \mathcal{D} \subseteq \mathcal{K}$, the pair $(\mathcal{C}, \mathcal{D})$ is a *strict factorisation system* of $\mathcal{K}$, written $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$, if every morphism $f \in \mathcal{K}$ factors *uniquely* as $f = c \circ d$ with $c \in \mathcal{C}$ and $d \in \mathcal{D}$. We call $\mathcal{K}$ the *Zappa–Szép product* of $\mathcal{C}$ and $\mathcal{D}$.

We fix the orientation once and for all: the $\mathcal{D}$ -part acts first (at the source), the $\mathcal{C}$ -part second (at the target). This is the many-object generalisation of the classical object. For monoids M, N (one-object categories), a Zappa–Szép product $M \bowtie N$ is a monoid K containing M, N as submonoids for which the multiplication map $M \times N \rightarrow K$, $(m, n) \mapsto mn$, is a bijection: every element has a unique expression as an M -element times an N -element. This is the maximally general way two monoids sit inside a third; when N is normal it is the semidirect product $M \ltimes N$, when both are normal the direct product [36].

The point that makes the weld nontrivial with many objects is that $\mathcal{D}$ need *not* be the endomorphisms. It may carry *cross-object* arrows, so a factorisation $f = c \circ d$ moves the object at which the two factors meet. Welding then demands that a single “pivot object” be located, simultaneously and coherently, for every morphism of $\mathcal{K}$. This is a genuine recognition problem with no one-object shadow: at one object a factorisation either exists and one writes it down, or it does not.

5.2 The distributive law: the algebra of the weld

When a weld exists, $\mathcal{K}$ is reconstructed from $\mathcal{C}$, $\mathcal{D}$, and a distributive law $\lambda: \mathcal{D} \triangleleft \mathcal{C} \Rightarrow \mathcal{C} \triangleleft \mathcal{D}$ generalising the matched-pair calculus of monoids to two genuine many-object categories. This is the categorical form of the correspondence between distributive laws and strict factorisations [37, 38], and for directed containers it is exactly Ahman–Uustalu’s composition construction [27]. We record the data, the axioms, and the associativity theorem.

Throughout, for composable $d \in \mathcal{D}$ and $c \in \mathcal{C}$ we write ${}^d c$ for the *left action* of d on c (a $\mathcal{C}$ -arrow) and c^d for the *right restriction* of d by c (a $\mathcal{D}$ -arrow).

Definition 5.2 (Distributive law / matched pair of categories). A *distributive law* $\lambda: \mathcal{D} \triangleleft \mathcal{C} \Rightarrow \mathcal{C} \triangleleft \mathcal{D}$ assigns to each composable pair (d, c) with $d \in \mathcal{D}$, $c \in \mathcal{C}$, $\text{dom}(d) = \text{cod}(c)$ a pair $\lambda(d, c) = ({}^d c, c^d)$ with ${}^d c \in \mathcal{C}$, $c^d \in \mathcal{D}$ and

$$\text{dom}(c^d) = \text{dom}(c), \quad \text{cod}({}^d c) = \text{cod}(d), \quad \text{dom}({}^d c) = \text{cod}(c^d),$$

so that ${}^d c \circ c^d$ has the same source and target as $d \circ c$. It is a *matched pair* if, for all composable data,

$$(U) \quad \text{id}^c c = c, \quad c^{\text{id}} = \text{id}_{\text{dom } c}, \quad {}^d \text{id} = \text{id}_{\text{cod } d}, \quad \text{id}^d = d;$$

$$(ZS2) \quad ({}^{d' \circ d} c) = d'({}^d c);$$

$$(ZS3) \quad c^{(d' \circ d)} = ({}^d c)^{d'} \circ c^d;$$

$$(ZS1) \quad {}^d(c_2 \circ c_1) = {}^d c_2 \circ ({}^{c_2^d} c_1);$$

$$(ZS4) \quad (c_2 \circ c_1)^d = c_1^{(c_2^d)}.$$

The four axioms are Brin’s matched-pair axioms [36], typed for two many-object categories: (ZS2) says $\mathcal{D}$ acts on $\mathcal{C}$ on the left, (ZS3) is the cocycle for the restriction over $\mathcal{D}$ -composition, and (ZS1),(ZS4) are the compatibilities with $\mathcal{C}$ -composition. They reduce verbatim to the monoid axioms when each factor has one object.

Definition 5.3 (The product category). Given a matched pair, $\mathcal{C} \bowtie \mathcal{D}$ has the shared object set, hom-sets $(\mathcal{C} \bowtie \mathcal{D})(a, b) = \coprod_m \mathcal{C}(m, b) \times \mathcal{D}(a, m)$, identities $(\text{id}_a, \text{id}_a)$, and composition

$$(c_2, d_2) \cdot (c_1, d_1) = (c_2 \circ {}^{d_2} c_1, c_1^{d_2} \circ d_1),$$

for $(c_1, d_1): a \rightarrow b$ (so $d_1: a \rightarrow m_1$, $c_1: m_1 \rightarrow b$) and $(c_2, d_2): b \rightarrow e$.

At one object this is precisely the classical Zappa–Szép multiplication $(m_2, n_2)(m_1, n_1) = (m_2 {}^{n_2} m_1, m_1^{n_2} n_1)$.

Theorem 5.4 (Axioms $\iff$ associativity). (proved) $\mathcal{C} \bowtie \mathcal{D}$ of Definition 5.3 is a category if and only if λ is a matched pair. The unit laws are equivalent to (U); a single associativity instance splits into its $\mathcal{C}$ -coordinate, governed by (ZS1)+(ZS2), and its $\mathcal{D}$ -coordinate, governed by (ZS3)+(ZS4).

Proof sketch. The unit laws for (id, id) unwind directly to (U). For associativity, expand both bracketings of $(c_3, d_3)(c_2, d_2)(c_1, d_1)$ by Definition 5.3. Both have a $\mathcal{C}$ -composite first coordinate and a $\mathcal{D}$ -composite second coordinate, and the two expansions agree iff the coordinates agree separately. The $\mathcal{C}$ -coordinate discrepancy is $^{(d_3 \circ d_2)}c_1$ versus $d_3(^{d_2}c_1)$ together with a distribution of $^d(-)$ over a $\mathcal{C}$ -composite — exactly (ZS2) and (ZS1). The $\mathcal{D}$ -coordinate discrepancy is $c_1^{(d_3 \circ d_2)}$ versus $(^{d_2}c_1)^{d_3 \circ c_1^{d_2}}$ together with the distribution of $(-)^d$ over a $\mathcal{C}$ -composite — exactly (ZS3) and (ZS4). The only place $\mathcal{D}$ being a genuine category rather than a family of monoids enters is that the restrictions composed in the $\mathcal{D}$ -coordinate must be composable across a change of object; the typing of Definition 5.2 guarantees this. The converse reads the same identity chain from associativity back to the axioms. $\square$

The next lemma runs the correspondence the other way: it extracts the law from a weld and certifies that the abstract product recovers $\mathcal{K}$.

Lemma 5.5 (Extraction from a strict factorisation). *Let $(\mathcal{C}, \mathcal{D})$ be a strict factorisation of $\mathcal{K}$. For composable $d \in \mathcal{D}$, $c \in \mathcal{C}$ let $^d c \in \mathcal{C}$, $c^d \in \mathcal{D}$ be the unique factorisation of $d \circ c \in \mathcal{K}$:*

$$d \circ c = ^d c \circ c^d.$$

Then λ is a matched pair, and $\Phi: \mathcal{C} \bowtie \mathcal{D} \rightarrow \mathcal{K}$, $\Phi(c, d) = c \circ d$, is an isomorphism of categories.

Proof sketch. Each axiom is obtained by factoring one $\mathcal{K}$ -morphism two ways and invoking uniqueness. For (ZS2)/(ZS3), factor $(d' \circ d) \circ c$ as

$$(d' \circ d) \circ c = d' \circ (^d c \circ c^d) = (d' \circ ^d c) \circ c^d = d'(^d c) \circ (^d c)^{d'} \circ c^d,$$

the last step factoring $d' \circ ^d c$, composable since $\text{cod } ^d c = \text{cod } d = \text{dom } d'$; here $(^d c)^{d'}$ and c^d compose in $\mathcal{D}$ because $\text{dom } (^d c)^{d'} = \text{dom } ^d c = \text{cod } c^d$ — the one place $\mathcal{D}$ being a genuine category matters. Comparing with the direct factorisation of $(d' \circ d) \circ c$ and invoking uniqueness yields the $\mathcal{C}$ -part (ZS2) and $\mathcal{D}$ -part (ZS3). Factoring $d \circ (c_2 \circ c_1)$ two ways yields (ZS1)/(ZS4), and the units are immediate. Finally Φ is a bijection on each hom-set (it *is* the factorisation), preserves identities, and preserves composition by the rewrite $d \circ c = ^d c \circ c^d$, so it is an isomorphism. $\square$

Thus when a weld exists, $\mathcal{K}$ is genuinely the Zappa–Szép product of the categories $\mathcal{C}$ and $\mathcal{D}$, and λ is the precise carrier of the mutual action — the data one would extract to *run* the composite system.

5.3 The recognition problem and the free-presheaf lemma

Theorem 5.4 and Lemma 5.5 describe the algebra *once* a weld exists. The decision problem is prior: given $\mathcal{K}$ and a prescribed right factor $\mathcal{D}$, does a complementary $\mathcal{C}$ exist at all? The whole answer turns on one observation — the presheaf form of “free module = has a basis” — which we isolate first.

The hom-sets of $\mathcal{K}$ into a fixed target b assemble into a presheaf: a right $\mathcal{D}$ -set.

Definition 5.6 (Right $\mathcal{D}$ -set, representable, free, target presheaf). A *right $\mathcal{D}$ -set* is a presheaf $M: \mathcal{D}^{\text{op}} \rightarrow \mathbf{Set}$: each $d: a \rightarrow m$ gives $x \mapsto x \cdot d$, with $x \cdot \text{id} = x$ and $(x \cdot d') \cdot d = x \cdot (d' \circ d)$. The *representable* at m is $\mathcal{D}(-, m)$ with action by precomposition; M is *free* if it is isomorphic to a coproduct of representables. For an object b of $\mathcal{K}$, the *target presheaf* $\text{Hom}_{\mathcal{K}}(-, b): \mathcal{D}^{\text{op}} \rightarrow \mathbf{Set}$ sends a to $\text{Hom}_{\mathcal{K}}(a, b)$ and $d: a' \rightarrow a$ to precomposition $f \mapsto f \circ d$; it is a right $\mathcal{D}$ -set because $\mathcal{D} \subseteq \mathcal{K}$.

Lemma 5.7 (Free = unique factorisation). (proved) *Let $\mathcal{D}$ be a small category and $M: \mathcal{D}^{\text{op}} \rightarrow \mathbf{Set}$. The following are equivalent.*

1. *M is free.*
2. *There is a family $B = (B_m \subseteq M(m))_{m \in \text{ob } \mathcal{D}}$ such that for every object a and every $x \in M(a)$ there is a unique pair (c, d) with $c \in B_m$ (some m) and $d \in \mathcal{D}(a, m)$ satisfying $x = c \cdot d$.*

When they hold, $M \cong \coprod_m \coprod_{c \in B_m} \mathcal{D}(-, m)$, and B is a basis.

Proof sketch. (2) $\Rightarrow$ (1): $\varphi_a(c, d) = c \cdot d$ is a well-defined presheaf morphism by contravariance of M and precomposition on $\mathcal{D}(-, m)$; it is objectwise bijective by hypothesis, hence an isomorphism. (1) $\Rightarrow$ (2): given $\psi: \coprod_i \mathcal{D}(-, m_i) \cong M$, set $c_i := \psi_{m_i}(\text{id}_{m_i})$; by Yoneda the i -th component is “multiply by c_i ”, and since ψ is an isomorphism every x equals $c_i \cdot d$ for a unique index i and — the representable being a genuine set of morphisms — a unique d . This is (2) with $B_m = \{c_i : m_i = m\}$. $\square$

Remark 5.8 (The endomorphism specialisation). If $\mathcal{D} = \mathcal{E}$ is the disjoint union of vertex monoids ($\mathcal{E}(a, a) = \text{End}(a)$, $\mathcal{E}(a, b) = \emptyset$ for $a \neq b$), a presheaf on $\mathcal{E}$ is a family of right $\text{End}(a)$ -sets, freeness is freeness of each fibre, and a basis B_a is one representative per $\text{End}(a)$ -orbit. So Lemma 5.7 reduces to “free $\text{End}(a)$ -set = has an orbit transversal” fibre by fibre.

5.4 The pairwise criterion

The key observation is that, for a weld, a free basis of $\text{Hom}_{\mathcal{K}}(-, b)$ is *forced* to be the family of $\mathcal{C}$ -arrows into b . There is nothing to choose: the factorisation dictates the basis and the basis dictates the factorisation.

Theorem 5.9 (Pairwise Zappa–Szép criterion). (proved) *Let $\mathcal{K}$ be a small category and $\mathcal{D} \subseteq \mathcal{K}$ a wide subcategory. The following are equivalent.*

1. *There is a wide subcategory $\mathcal{C} \subseteq \mathcal{K}$ with $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$.*
2. *Both:*

(L) *for every object b , the right $\mathcal{D}$ -set $\text{Hom}_{\mathcal{K}}(-, b)$ is free; and*

(G) *free bases B_b of $\text{Hom}_{\mathcal{K}}(-, b)$ can be chosen so that $\mathcal{C} := \bigcup_b B_b$ is a wide subcategory of $\mathcal{K}$.*

Proof sketch. The only interaction the proof uses is that a basis element $c \in B_m \subseteq \text{Hom}_{\mathcal{K}}(m, b)$ and $d \in \mathcal{D}(a, m)$ give the composite $c \cdot d = c \circ d: a \rightarrow b$ in $\mathcal{K}$.

(1) $\Rightarrow$ (2): with $B_b := \{c \in \mathcal{C} : \text{cod}(c) = b\}$, uniqueness of the factorisation says each $f: a \rightarrow b$ is $c \circ d$ for a unique $c \in B_b$, $d \in \mathcal{D}(a, \text{dom } c)$; by Lemma 5.7, B_b is a basis, so $\text{Hom}_{\mathcal{K}}(-, b)$ is free — this is (L). Since each $\mathcal{C}$ -arrow has one codomain, $\bigcup_b B_b = \mathcal{C}$, wide by hypothesis — this is (G).

(2) $\Rightarrow$ (1): let $\mathcal{C} = \bigcup_b B_b$ be the wide subcategory from (G); then $B_b = \{c \in \mathcal{C} : \text{cod}(c) = b\}$. Freeness with basis B_b gives, for each $f: a \rightarrow b$, a unique (c, d) with $c \in B_b \subseteq \mathcal{C}$, $d \in \mathcal{D}$, $f = c \circ d$. Any $f = c' \circ d'$ with $c' \in \mathcal{C}$ has $\text{cod}(c') = b$, so $c' \in B_b$, and uniqueness over B_b forces $(c', d') = (c, d)$. Hence the factorisation is unique over all of $\mathcal{C}$, and $(\mathcal{C}, \mathcal{D})$ is a strict factorisation. $\square$

Remark 5.10 (Why two levels). Condition (L) concerns the right $\mathcal{D}$ -action alone: it asks that the action be *free*, so bases exist, one choice per target. It is the analogue of “free module,” and is checkable target by target. Condition (G) is the cross-target constraint: the independently chosen bases must *cohere* into a single subcategory. The two can come apart (Proposition 5.15), and the gap is a holonomy — the failure of the candidate λ to be consistently single-valued. Asking for (G) is exactly asking when a wide $\mathcal{D}$ *extends* to a strict factorisation, the subject of Rosebrugh–Wood [37]. Its cohomological classification is the business of the next chapter (Chapter 6); here we record only the shape of the obstruction.

The single-category specialisation

Corollary 5.11. (proved) *Take $\mathcal{D} = \mathcal{E}$, the endomorphism subcategory. Then $\mathcal{K} = \mathcal{C} \bowtie \mathcal{E}$ for some wide $\mathcal{C}$ if and only if (L') each $\text{Hom}_{\mathcal{K}}(a, b)$ is free as a right $\text{End}(a)$ -set, and (G') per-orbit representatives can be chosen to close into a wide subcategory.*

Proof sketch. By Remark 5.8, freeness of the $\mathcal{E}$ -presheaf $\text{Hom}_{\mathcal{K}}(-, b)$ is fibrewise freeness of each $\text{Hom}_{\mathcal{K}}(a, b)$ over $\text{End}(a)$, a basis is one representative per orbit, and (G) becomes closure of the representatives. $\square$

The classical anchor

At one object the recognition problem is vacuous and only the algebra survives.

Proposition 5.12 (Classical Zappa–Szép recovered). (proved) *Let M, N be monoids as one-object categories. A matched pair (Definition 5.2) is exactly a left action ${}^n m$ of N on M and a right action m^n of M on N satisfying Brin’s four axioms, and the product $M \bowtie N$ (Definition 5.3) is the classical Zappa–Szép product with multiplication $(m_2, n_2)(m_1, n_1) = (m_2 {}^{n_2} m_1, m_1^{n_2} n_1)$. Such products are precisely the monoids K with submonoids M, N for which $M \times N \rightarrow K$ is a bijection.*

Proof sketch. With one object every typing constraint in Definition 5.2 is vacuous, (U) and (ZS1)–(ZS4) become the unit laws and Brin’s four identities [36], and the coproduct $\coprod_m$ in Definition 5.3 is trivial. The last sentence is Theorem 5.9 read at one object: (L) is freeness of K as a right N -set and (G), for basis M , is that $M \times N \rightarrow K$ is a bijection with M a submonoid. $\square$

Remark 5.13 (Update monads, honestly). It is tempting to call Ahman–Uustalu’s update monad [28] the degenerate (semidirect) instance, but this overclaims and we avoid it. The update monad $TX = S \rightarrow (P \times X)$ is the writer-over-reader twist of a *monoid and a set* through a single one-sided action [28]; its directed container is the action category $S \rtimes P$ [26]. A Zappa–Szép product is the twist of two *monoids* through two mutually twisting actions. The honest relationship is analogy and handedness: the update monad is not literally a Zappa–Szép product of two monoids, and a genuine factorisation enters only if P itself splits as $P = A \bowtie B$.

5.5 A genuinely pairwise obstruction

One might suspect the two-level phenomenon is an artefact of $\mathcal{D}$ being the endomorphisms — that allowing cross-object arrows in $\mathcal{D}$ always dissolves the global obstruction. It does not.

Example 5.14 (Rigid twist over a cross-object factor). Let $\mathcal{K}$ have objects w, a, x, y with $\text{End}(a) = \{\text{id}_a, g\}$, $g^2 = \text{id}_a$, the other endomorphism monoids trivial, and

$$\text{Hom}(a, x) = \{p, pg\}, \text{Hom}(x, y) = \{s, s_2\}, \text{Hom}(a, y) = \{q, qg\}, \text{Hom}(w, a) = \{d, gd\},$$

$\text{Hom}(w, x) = \{pd, pgd\}$, $\text{Hom}(w, y) = \{qd, qgd\}$, where $pg = p \circ g$, $gd = g \circ d$, the rigid twist $s \circ p = q$, $s_2 \circ p = qg$, and the forced composites $p \circ d = pd$, $s \circ pd = qd$, $s_2 \circ pd = qgd$, etc. Let $\mathcal{D} = \{\text{id}_w, \text{id}_a, \text{id}_x, \text{id}_y, g, d, gd\}$.

Proposition 5.15. (proved) *In Example 5.14, $\mathcal{D}$ is a wide subcategory containing the cross-object arrows $d, gd: w \rightarrow a$; every $\text{Hom}_{\mathcal{K}}(-, b)$ is free as a right $\mathcal{D}$ -set (L holds), yet no wide $\mathcal{C}$ makes $(\mathcal{C}, \mathcal{D})$ a strict factorisation (G fails). Hence $\mathcal{K}$ has no weld over this $\mathcal{D}$ despite the cross-object factor.*

Proof sketch. $\mathcal{D}$ is closed ($g \circ g = \text{id}_a$, $g \circ d = gd$, $g \circ gd = d$). The bases $B_w = \{\text{id}_w\}$, $B_a = \{\text{id}_a\}$ (the orbit $\{\text{id}_a, g, d, gd\}$ is a single free $\mathcal{D}$ -orbit), $B_x = \{\text{id}_x, p\}$, $B_y = \{\text{id}_y, q, s, s_2\}$ witness (L); note s, s_2 are forced into B_y because $\mathcal{D}$ has no non-identity arrow out of x . For (G), any closing $\mathcal{C}$ must pick $t \in \{p, pg\}$ and a basis B_y containing s, s_2 and exactly one of q, qg . Closure needs $s \circ t$, $s_2 \circ t \in B_y$; but $s \circ p = q$ and $s_2 \circ p = qg$ (and symmetrically for pg) force B_y to contain *both* q and qg — impossible. So no $\mathcal{C}$ closes: (G) fails, and by Theorem 5.9 there is no weld. $\square$

Remark 5.16. The cross-arrow d enters $\mathcal{D}$ as a free $\langle g \rangle$ -torsor $\{d, gd\}$ into a ; it enlarges $\mathcal{D}$ to a genuine many-object category and keeps every hom-presheaf free, but it is inert against the branch obstruction at $a \rightarrow x \rightrightarrows y$. Promoting $\mathcal{D}$ from vertex monoids to a category with cross-object arrows does not dissolve the holonomy. This is Theorem 5.9 in its genuinely pairwise form. All claims of this section are machine-checked by finite-category scripts: exhaustive over all 1152 wide subcategories, $\mathcal{K}$ admits no strict factorisation over $\mathcal{D}$; and Theorem 5.4 is confirmed with zero disagreements over all type-correct laws in the tested finite cases.

5.6 The bridge: orchestration composites are Zappa–Szép products

The abstract weld becomes the operative composition tool through one further identification, which is where the container model earns the machinery above. In the polynomial-functor model of interaction [31], an agent that may issue queries from a set S and, for each s , receive a reply from $P(s)$ is the container $p = (S, P) = \sum_{s \in S} y^{P(s)}$; a handoff topology — who may delegate to whom, staying put, concatenating handoffs — is a directed container, i.e. a small category (Definition 3.8). Running *two* orchestrations that share a resource is the interleaving of two directed containers, and this interleaving is a Zappa–Szép product $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$.

Proposition 5.17 (Holonomy–composition / Zappa–Szép bridge). (peer-reviewed) *The composite of two shared-resource orchestrations, presented as a small (handoff) category $\mathcal{K}$ with prescribed right factor $\mathcal{D}$ the resource-internal part, is the Zappa–Szép product $\mathcal{C} \bowtie \mathcal{D}$ of directed containers in the sense of Ahman–Uustalu’s distributive-law construction [27], with $\mathcal{C}$ the genuine dispatch/handoff factor. Consequently the joint agent exists if and only if the weld exists, i.e. iff conditions (L) and (G) of Theorem 5.9 both hold; when it does, the mutual action is carried by the distributive law λ of Definition 5.2.*

The identification itself — that the orchestration composite is $\mathcal{C} \bowtie \mathcal{D}$, and hence that its existence is governed by the weld — is the delta over Ahman–Uustalu, whose account builds the joint container when a distributive law is supplied but offers no measure of its absence. Proposition 5.17 is peer-reviewed (Rick); we take Ahman–Uustalu’s construction [27] as prior art and claim none of it. The concrete consequence is a decision procedure: a cheap fibrewise freeness test (L), checkable target by target, gates the more expensive global closure search (G), and only its successes feed the search. The matched-pair calculus of §5.2 then exhibits λ explicitly as the data one extracts to run the composite.

What remains is to make the residual obstruction (G) precise. It is a holonomy: the independently chosen free bases fail to glue, and the failure is a monodromy around the dispatch/return loop — the categorical form of “a flat bundle is trivial iff its holonomy vanishes.” Under the abelian-vertex, trivial-left-action hypothesis the free $\mathcal{D}$ -orbits form a base category, the vertex automorphism groups form a coefficient presheaf, and the defect of a transversal is a normalised 2-cocycle. The classification of (G) as a class in H^2 , and the precise statement that the weld exists iff $[\omega] = 0$, is the subject of the next chapter (Chapter 6); the cohomology conventions it uses are those of §3.5. Here we hand off (G) as exactly that: a global closure condition whose obstruction is a holonomy, waiting to be named.

Chapter 6

The correctness criterion

$$(L) \wedge [\omega] = 0 \in H^2$$

Here the two threads join into the thesis equation. The global condition (G) of the pairwise criterion is identified with the vanishing of a degree-two cohomology class $[\omega] \in H^2$, so that a composite is correct exactly when the local condition holds and the weld obstruction vanishes. The chapter also records the “two ω -sites” result: the degree-1 (prunability / descent) and degree-2 (holonomy / Schreier) obstructions live at genuinely different sites and must not be conflated.

6.1 The thesis equation

The preceding chapter developed the pairwise Zappa–Szép criterion: given a handoff category $\mathcal{K}$ (a directed container, Definition 3.8, under the chain identification $\mathbf{DCont} \cong \mathbf{Cat}$, Theorem 3.9) and a prescribed wide subcategory $\mathcal{D} \subseteq \mathcal{K}$, a complementary factor $\mathcal{C}$ making $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ a Zappa–Szép product exists iff two conditions hold — a *local* freeness condition (L) and a *global* closure condition (G) (see Chapter 5, from which we *receive* (L) and (G) and do not re-derive them). This chapter contributes the identification of the second, global condition with a cohomological vanishing. Writing $\mathrm{Sk}_{\mathcal{C}}$ for the skeleton orbit category and $\mathcal{D}$ for the reply (direction) functor as coefficients (cohomology conventions and the class $[\omega]$ as an LHS transgression: §3.5), the result of the chapter is that the global condition *is* a degree-two obstruction, and hence the whole correctness verdict collapses to a single displayed equivalence:

$$\boxed{\text{a composite is correct} \iff (L) \wedge [\omega] = 0 \in H^2} \tag{6.1}$$

Everything below either proves the right-hand identification (G) $\Leftrightarrow [\omega] = 0$, exhibits the smallest faithful instance where the class is machine-checked, or warns against conflating this degree-two obstruction with the genuinely distinct degree-one obstruction living at another site.

6.2 The global condition is a degree-two class

The bridge from a distributive law to a Zappa–Szép product is classical and is used here as an input, not proved: a strict factorization system $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ is equivalently a distributive law $\lambda : \mathcal{DC} \Rightarrow \mathcal{CD}$ satisfying the matched-pair axioms, the directed-container instance of which is the composition construction of Ahman–Uustalu (*peer-reviewed*). What the source supplies *over* that construction

is a measure of its failure: Ahman–Uustalu’s account builds the composite whenever a distributive law exists, but carries no invariant for when one does not.

Theorem 6.1 ((G) is the vanishing of $[\omega]$; (proved)). *Fix a handoff category $\mathcal{K}$ and a prescribed wide subcategory $\mathcal{D}$ for which the local condition (L) holds — each hom-presheaf is a free right $\mathcal{D}$ -set — and for which the vertex-group hypothesis holds ($\mathcal{D}$ a disjoint union of abelian vertex groups with trivial left action). Then the complementary factor exists iff a single degree-two class vanishes:*

$$\mathcal{C} \bowtie \mathcal{D} \text{ exists} \iff [\omega] = 0 \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}),$$

cohomology of the skeleton of the handoff category with coefficients in the direction functor $\mathcal{D}$ [17, 13]. Equivalently, the global condition (G) of the pairwise criterion is exactly the statement $[\omega] = 0$.

The cohomology itself — Baues–Wirsching cohomology of a small category, and its appearance in Schreier/factorization theory — is classical; the delta over the peer-reviewed construction is the *identification* of the orchestration composite with $\mathcal{C} \bowtie \mathcal{D}$ and hence of its obstruction with this class. Combining Theorem 6.1 with the received pairwise criterion of Chapter 5 yields the boxed equation (6.1) directly: correctness is $(L) \wedge (G)$, and (G) is $[\omega] = 0$.

Remark 6.2 (reading the class as holonomy). The class is the categorical form of “a flat bundle is trivial iff its holonomy vanishes”. Under (L) each partial handoff completes freely; the completions fail to glue into a single serialized joint agent exactly by the monodromy $[\omega]$ around the dispatch/return loop. Thus (L) is completability of the parts and $[\omega] = 0$ is coherence of their gluing — the two halves of (6.1).

6.3 The minimal faithful instance: $[\omega] = \varepsilon$, machine-checked

The smallest place where the answer is “no” is a supervisor dispatching to a worker that may call back before returning — unprotected re-entrancy, the DAO-class failure. This instance is faithful and its class is computed on the nose (*lean-verified*).

Definition 6.3 (the parametrized supervisor–worker category). Fix roles $\mathbf{S}$ (supervisor), $\mathbf{W}$ (worker), $\mathbf{R}$ (return), with the supervisor carrying a one-bit turn token, so $\text{End}(\mathbf{S}) = \{\text{id}, \tau\} \cong \mathbb{Z}/2$. For $\varepsilon \in \mathbb{Z}/2$ let $\mathcal{K}_{\varepsilon}$ have generating handoffs $p, p\tau : \mathbf{S} \rightarrow \mathbf{W}$ and worker outcomes $s, s_2 : \mathbf{W} \rightarrow \mathbf{R}$, with the single length-two composite fixed by

$$s \circ p = q, \quad s_2 \circ p = q \cdot \tau^{\varepsilon}.$$

The prescribed right factor is the token part $\mathcal{D} = \{\text{id}_{\mathbf{S}}, \tau, \text{id}_{\mathbf{W}}, \text{id}_{\mathbf{R}}\}$. The bit ε records whether the re-entrant outcome s_2 *mutates* the token ($\varepsilon = 1$) or *fixes* it ($\varepsilon = 0$, a lock).

The categorical obstruction of Theorem 6.1 reduces, for $\mathcal{K}_{\varepsilon}$, to a finite $\mathbb{F}_2$ cochain complex. The base $\text{Sk}_{\mathcal{C}}$ is the branch $\mathbf{S} \rightarrow \mathbf{W} \rightrightarrows \mathbf{R}$; the normalized (Baues–Wirsching) complex has $C^1 \cong \mathbb{F}_2^2$, $C^2 \cong \mathbb{F}_2^2$, $C^3 = 0$, so every 2-cochain is a cocycle. The coboundary image is the diagonal $B^2 = \langle (1, 1) \rangle$, whence $H^2 = C^2/B^2 \cong \mathbb{F}_2$, realised by the class map $\varphi(a, b) = b - a = a \oplus b$ with $\ker \varphi = B^2$. The defect cocycle of the natural transversal is $\omega_{\varepsilon} = (0, \varepsilon)$, so $\varphi(\omega_{\varepsilon}) = \varepsilon$.

Theorem 6.4 ($[\omega] = \varepsilon$; (*lean-verified*)). *In a sorry-free Lean 4 development using no external library (Containers/Reentrancy.lean), the following are machine-checked over the finite complex above:*

- $\ker \varphi = B^2$ and φ surjective, together giving $H^2 = C^2/B^2 \cong \mathbb{F}_2$;

- $\varphi(\omega_\varepsilon) = \varepsilon$ — the class form of $[\omega] = \varepsilon$;
- $\omega_\varepsilon \in B^2 \iff \varepsilon = 0$ — the membership form of $[\omega] = 0 \iff \varepsilon = 0$.

Consequently, on the minimal faithful model, the joint agent $\mathcal{C} \bowtie \mathcal{D}$ exists iff the worker fixes the token, and unprotected re-entrancy ($\varepsilon = 1$) is the nonzero generator of $H^2 \cong \mathbb{Z}/2$.

Remark 6.5 (the verification boundary, drawn honestly). The Lean development certifies the *finite $\mathbb{F}_2$ class computation* — the last mile, from cochain complex to $[\omega] = \varepsilon$. The *reduction* of the categorical obstruction of Theorem 6.1 to that complex — that $\mathcal{K}_\varepsilon$ satisfies (L) and the vertex-group hypothesis, that $\text{Sk}_\mathcal{C}$ is the stated branch, and that the transversal defect is $(0, \varepsilon)$ — is an (*proved*) pen-and-paper input, not formalised. So: the last mile is machine-checked; the first mile is a cited pen-and-paper theorem.

6.4 Two ω -sites: degree-1 descent versus degree-2 Schreier

The chapter closes with a warning that is also the natural lead-in to the following chapter. It is tempting to read *every* obstruction to composing directed containers as the class $[\omega]$ of (6.1). This is false: there are two genuinely different obstruction sites, at two different cohomological degrees, and they must not be conflated (*proved*).

Theorem 6.6 (the two ω -sites are irreducibly distinct; (*proved*)). *The obstruction to composing directed containers factors through two independent sites:*

- a **degree-1** (*prunability / descent*) obstruction — the degree-one kernel-descent class, Ferri’s condition on the ideal generated by the weld; and
- the **degree-2** (*holonomy / Schreier*) obstruction $[\omega] \in H^2$ of Theorem 6.1 — the Zappa–Szép monodromy.

These live at genuinely different sites [16, 14, 15, 18]: *prunability is a descent condition, not a Schreier H^2 class, and the two decouple — there are witnesses with $[\omega] = 0$ yet nontrivial degree-one obstruction, and conversely. In particular, “prunability fails” and “ $[\omega] \neq 0$ ” are not the same verdict, and identifying them is a category error.*

Concretely, the degree-1 obstruction is the descent/subfunctor condition governing whether a weld can be *pruned* to a subobject, while the degree-2 class $[\omega]$ governs whether the parts glue into a global serialized agent at all; on suitable witnesses one vanishes while the other does not, so neither dominates the other.

Remark 6.7 (a retired side-reading). An earlier framing predicted that a “total” versus “partial” composition split would be separated by a degree-one invariant. That specific prediction is withdrawn: for the correctness verdict (6.1) the single degree-two class $[\omega]$ is the whole invariant, both the obstruction to existence and, via $H^2 \cong \mathbb{F}_2$ on the minimal model, the label of which system one has. The genuine two-site phenomenon of Theorem 6.6 is a different statement — prunability/descent versus Schreier holonomy — and is not the retired split.

The degree-2 class $[\omega]$ is, moreover, only *one* of three obstruction theories that live on $\mathbf{DCont} \cong \mathbf{Cat}$: the Zappa–Szép holonomy of this chapter, the prunability/descent condition of Theorem 6.6, and a third category-cohomological home. The comparison of all three, and the question of a unifying protomodular home, is the subject of the following chapter 7; the two- ω -sites result above is the entry point to it.

Chapter 7

The H^2 -cluster of obstruction theories

The previous chapter distilled correctness to a single class, $[\omega] = 0 \in H^2$. That is one obstruction theory on the object $\mathbf{DCont} \simeq \mathbf{Cat}$. It is not the only one. Once the Zappa–Szép holonomy is in hand, two further degree-two obstruction theories present themselves on the *same* object — Ferri’s prunability obstruction and the direction-functor cohomology $H^2_{\mathbf{Cat}_B}$ — and the natural reflex is to hope they are three costumes of one theorem, three faces glued by a single universal H^2 . They are not. What this chapter records is the honest shape: a *structured family* of three genuinely distinct theories, with one proved separation between them, one refuted unifier (a valuable negative), and a single organising grammar that survives the refutation — the cluster is a *filtration*, not a product.

7.1 Three homes on one object

(a) **The multiplicative holonomy** $[\omega]$. The Zappa–Szép composition holonomy of Chapters 4–5 is a degree-two class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ over the orbit base $\mathrm{Sk}_{\mathcal{C}}$ (the skeleton of $\mathcal{C}$), with the criterion $(G) \iff [\omega] = 0$ (*proved*). It is a *multiplicative* invariant: it obstructs the existence of a coherent multiplicative transversal $\mathcal{C} \bowtie \mathcal{D}$. Two features fix its character. It vanishes for the *full loop bundle* and, more generally, on any *connected* groupoid, where the orbit base is coarse ($\simeq *$) and $H^{\geq 1}$ is forced to zero; it is nonzero only for a *proper* transversal $\mathcal{D} \subsetneq \pi$, where $\mathrm{Sk}_{\mathcal{C}} = B\Gamma$ carries the Schreier class of the extension $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$ (nonzero already for $\mathbb{Z}/4$ over $\mathbb{Z}/2$) (*proved*). The degree-two computations underneath it are backed by a machine-checked engine: the cyclic and fibre H^2 modules are *lean-verified*, and the emergent-holonomy count is identified with Ext^0 together with a vanishing theorem killing the higher Ext-tower [17] (*proved; engine lean-verified*).

(b) **Ferri’s additive prunability**. Ferri’s prunability obstruction [18] asks whether the loop bundle G° of a quiver skew brace is an ideal — concretely, whether the loops over each object form an *additive* subgroup that the cross-action carries to loops. It is the *additive twin* of the multiplicative Schreier line of (a). Two facts pin its place in the cluster, and both are corrections of a tempting error.

First, it is *orthogonal* to $[\omega]$: on the full loop bundle the composition holonomy vanishes identically (Proposition A, *proved*) while prunability can still fail, so the two are separate obstructions — the existence of a connected, non-prunable witness that certifies the separation is *computed* [15]. Second, and sharper: prunability is *not* a degree-two class at all. In the untwisted core where pruning is even well posed, it is a *degree-one* datum — the condition that the loop subfunctor be $\dashv$ -flat, i.e. that G° be a *kernel* (a descent/parallel-transport condition), logically independent of the degree-two star-reassembly (H^2) class one degree above it [16] (*proved; the coset reformulation*).

proved, the Schreier-vs-descent crosstab computed). Reading Ferri’s non-prunability of $K_{2,3}$ as an “[ω] $\neq 0$ ” phenomenon is therefore a *category error (refuted)*: $K_{2,3}$ is connected, so [ω] = 0 there, and its obstruction lives one degree lower. *An ideal is a kernel; an extension is a reassembly*.

(c) **The direction-functor cohomology $H^2_{\mathbf{Cat}_B}$** . The third home is external machinery, not ours. Ambra, Duvieusart, and Montoli [19] equip the category $\mathbf{Cat}_B$ of internal categories with a cohomology relative to *Schreier points*: $\mathbf{Cat}_B$ is $\mathcal{S}$ -Mal’tsev there, and H^{n+1} is computed as π_0 of the fibres of a *direction functor*. For a connected groupoid this reduces to the vertex group, $H^2_{\mathbf{Cat}_B}(G; A) \cong H^2(\pi; M)$ (*proved; the external machinery deep-read*), placing $H^2_{\mathbf{Cat}_B}$ on the *additive*/vertex axis alongside prunability and complementary to the (base-axis) [ω]. The authors draw *no* link to containers or Zappa–Szép products, so this is not a scoop of our programme; it is reusable, independently developed infrastructure that the cluster can borrow.

7.2 The refuted unifier, and the honest umbrella

The three homes invite a single question: is there one protomodular / internal-semidirect-product H^2 that merges the multiplicative [ω] and the additive prunability into faces of a common class? The answer is *no*, and the negative is part of the result (*refuted, 2026-10-09*).

The structural reason is a clean divide. A connected groupoid splits canonically as $G \cong I(\mathrm{ob} G) \times B\pi$ — coarse base times vertex group — and the two obstruction theories are the cohomologies of the two complementary factors. But these factors sit on opposite sides of the *pointed* / *non-pointed* boundary. The vertex factor $B\pi$ is pointed, so the protomodular direction-functor $H^2(\pi; M)$ of (c) applies to it and carries the additive face. The base factor $\mathrm{Sk}_C = I(\mathrm{ob} G)$ is *non-pointed* — it has no zero object — so no protomodular H^2 sees it at all; [ω] is a nerve/simplicial class, vanishing exactly when the base is connected. A single semi-abelian H^2 cannot straddle the divide, because protomodularity is exactly the structure the non-pointed factor lacks.

What replaces the hoped-for single class is a *filtration*. The honest umbrella is a Künneth / product-nerve spectral sequence, whose abutment $H^2_{\mathrm{cat}}(G; M)$ is filtered rather than split. The natural target is

$$H^2_{\mathrm{cat}}(G; M) \cong H^2(\pi; M) \oplus H^2_{\mathrm{base}}(\mathrm{Sk}_C; M), \quad (7.1)$$

with the direction functor realising the vertex-group projection. This is registered as an open node and it is *speculative* — provable from the $H^2_{\mathbf{Cat}_B}$ reduction and the separation in the connected / field-coefficient case, where the filtration collapses, with genuine content only in the disconnected / proper-transversal case, where it does not. The refuted unifier and the speculative filtration are two readings of one fact: the coupling between the base and vertex faces is a *differential*, not a direct sum.

That last sentence is not a metaphor. On the normal Zappa–Szép slice the categorical cohomology $H^2_{\mathrm{cat}}(G) \cong H^2(\pi; M)$ is the abutment of the Lyndon–Hochschild–Serre spectral sequence of $1 \rightarrow \mathcal{D} \rightarrow \pi \rightarrow \Gamma \rightarrow 1$, whose two degree-two edges are exactly the two faces — the base edge $H^2(\Gamma; M^{\mathcal{D}})$ carrying [ω], the fibre edge $H^2(\mathcal{D}; M)^{\Gamma}$ carrying the additive reassembly class — and whose transgression *is* the holonomy, $d_2(\varphi) = \varphi_*[\omega]$ (*proved*) [13]. The naive direct sum (7.1) holds *if and only if* [ω] = 0; whenever [ω] $\neq 0$ the base edge does not survive to the abutment (it is a d_2 -image, not a summand), so the two faces are never simultaneously present as independent summands. The unifier fails for precisely the reason the differential is nonzero.

7.3 The crown: the cluster is filtered, not a product

The same grammar recurs one level up, and this is the connection that turns the cluster from a list into a structure (*proved, 2026-09-23*). Ferri's decomposition $G \cong (G/G^\odot) \bowtie G^\odot$ has an obstruction that is a *two-stage tower*, not a sum of two independent classes [14]:

- **Tier 2 (degree ≤ 1):** *is $G^\odot$ a kernel?* This is the $\rightarrow$ -descent condition of (b) — for $|L_\lambda| = 2$ a single degree-one cochain — and it vanishes exactly when the quotient $G/G^\odot$ *exists*.
- **Tier 1 (degree 2):** *given the quotient, does the loop extension split?* This is the class $[\omega] \in H^2$ of (a) — which is *only defined once Tier 2 vanishes*, and which is automatically zero on connected G .

The tiers are *filtered*: Tier 1 is undefined until Tier 2 vanishes, because the group $H^2(\text{Sk}_C; \mathcal{D})$ that houses $[\omega]$ is built from a quotient and a coefficient system that do not exist until $G^\odot$ is a kernel. A higher differential lives only on the kernel of the lower one — exactly the shape of the LHS transgression $d_2(\varphi) = \varphi_*[\omega]$ of the previous section, now read at the level of the decomposition itself. On connected G Tier 1 is free, so the decomposition exists iff G is prunable, recovering Ferri's Corollary 4.8 with its exact cohomological anatomy; $K_{2,3}$ is the canonical instance with Tier 1 = 0 and Tier 2 $\neq 0$ (*proved*).

This is the whole point of assembling the cluster. The three obstruction theories are not three costumes of one theorem, and they are not merged by one universal class; the unifier is refuted. What binds them is a single organising grammar — *filtration, not product* — appearing twice: once inside the abutment (the transgression coupling the two edges) and once above it (the two-tier tower coupling descent to splitting). A degree-one datum must vanish before the degree-two datum is even defined, and a nonzero degree-two class is a differential, not a summand. The honest deliverable of this chapter is that grammar, together with the proved separation and the honestly labelled negative that forced it.

[*Assembly note: separation proved (existence witness computed); prunability-is-degree-1-descent proved; $H^2_{\text{Cat}_B}$ reduction proved; protomodular unifier refuted; product decomposition (7.1) speculative; two-tier tower proved; H^2 -engines lean-verified.*]

Chapter 8

Change-of-base, $\mathbf{Fam}(\mathcal{C}^{\mathrm{op}})$ generality, and the $\{\mathrm{Id}\}$ classification

How far does the equation generalise? The natural setting is containers over a general base, i.e. $\mathbf{Fam}(\mathcal{C}^{\mathrm{op}})$, with change-of-base along functors and monads. This chapter assembles the positive generality results (m -containers, codensity/polynomiality, the four monoidal structures) and the sharp negative: the self-enriched full-and-faithful change-of-base container semantics collapses to $\{\mathrm{Id}\}$ — no nontrivial effect monad supports it. Knowing the boundary is itself the result.

8.1 The base as a degree of freedom

Recall from Chapter 3 that a container is an object of $\mathbf{Fam}(\mathbf{Set}^{\mathrm{op}})$ (Notation 3.6) and that its extension is a left Kan extension along the representable embedding (Theorem 3.7),

$$\llbracket S, P \rrbracket X = \sum_{s \in S} \mathbf{Set}(P_s, X) = \sum_{s \in S} X^{P_s}.$$

The formula has exactly one silent parameter: the hom-set $\mathbf{Set}(P_s, X)$. Replace $\mathbf{Set}$ by a category $\mathcal{C}$ and read positions out in $\mathcal{C}$'s internal hom, and the same syntax presents a wholly different world.

Definition 8.1 (Containers over a base). For a closed symmetric monoidal category $(\mathcal{C}, \otimes, I, [-, -])$ with small coproducts, $\mathbf{Fam}(\mathcal{C}^{\mathrm{op}})$ has objects (S, P) with S a set of shapes and $P_s \in \mathcal{C}$ a *position object* for each s , and morphisms $(S, P) \rightarrow (T, Q)$ a shape map $f: S \rightarrow T$ together with, per s , a $\mathcal{C}$ -map $Q_{f(s)} \rightarrow P_s$ (contravariant in positions). The *extension* into $\mathcal{C}$ -enriched endofunctors is

$$\llbracket S, P \rrbracket = \sum_{s \in S} [P_s, -] : \mathcal{C} \rightarrow \mathcal{C}.$$

A single construction thus covers the whole programme, selected by the base: $\mathcal{C} = \mathbf{Set}$ gives ordinary containers; $\mathcal{C} = \mathbf{Kl}(M)$ for a monad M gives *effectful* (M -)containers, whose positions are read in a Kleisli category; $\mathcal{C} = \mathbf{Vec}$ gives linear containers; and $\mathcal{C} = \mathbf{Cat}$ recovers directed containers and hence small categories (Chapter 3, §3.3). The assignment $\mathcal{C} \mapsto \mathbf{Fam}(\mathcal{C}^{\mathrm{op}})$ is functorial: a functor $F: \mathcal{C} \rightarrow \mathcal{D}$ induces $\mathbf{Fam}(F^{\mathrm{op}})$, acting as F on positions and identically on shapes, and the total category assembling all bases is a bifibration over $\mathbf{Set}$ whose fibrewise opposite is the contravariance in positions [43]. *Change of base* along $F: \mathbf{Set} \rightarrow \mathcal{C}$ is the transport of the standard theory into $\mathcal{C}$; when F is (co)monoidal it carries (co)monoidal structure, and that is where effects and directedness enter.

The remainder of the chapter is a suite of positive generality results (§§8.2–8.5), a diagnosis of the linear wall (§8.6), and the collapse that fixes the exact boundary of a *uniform* enriched semantics (§8.7).

8.2 Positive suite I: m -containers, codensity, polynomiality

The Kleisli base is the central instance and the one where two Kan-theoretic invariants of the effect first appear. Fix a monad $M = (M, \eta, \mu)$ on **Set**; an M -container is a container (S, P) whose positions are read in $\mathbf{Kl}(M)$, with M -extension $\llbracket S, P \rrbracket_M X = \sum_s \mathbf{Set}(P_s, MX)$. All results of this section are *peer-reviewed* (Rick).

Theorem 8.2 (Identification and factorisation; *peer-reviewed*). $M\text{-}\mathbf{Cont} = \mathbf{Fam}(\mathbf{Kl}(M)^{\text{op}})$, and as endofunctors, naturally in (S, P) ,

$$\llbracket S, P \rrbracket_M = \llbracket S, P \rrbracket \circ M.$$

So an M -container functor is an *ordinary polynomial functor with a trailing M* . This one factorisation is the lever for everything below: every structural question about $\llbracket - \rrbracket_M$ becomes a question about the single trailing M . The proof is a direct unfolding of the two hom-descriptions.

Theorem 8.3 (Faithful always; fullness = codensity; *peer-reviewed*). $\llbracket - \rrbracket_M : M\text{-}\mathbf{Cont} \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is always faithful (the Kleisli unit η is a section, so container data is read back off any induced transformation). It is full — hence fully faithful — if and only if (i) each functor $\mathbf{Set}(A, M-)$ is connected, and (ii) M is codense, i.e. the canonical monad map $M \Rightarrow \text{Ran}_M M$ into its own codensity monad is an isomorphism. If M is affine ($M1 \cong 1$) then (i) holds automatically, so affine + codense $\Rightarrow$ fully faithful.

The engine is that precomposition $(-)\circ M$ has right adjoint Ran_M , so every hom-set $\mathbf{Nat}(\mathbf{Set}(A, M-), H)$ is a value of the right Kan extension along M ; the fullness obstruction is exactly the failure of the unit $M \Rightarrow \text{Ran}_M M = \Phi$ (the codensity monad, $\Phi(A) = \mathbf{Nat}(\mathbf{Set}(A, M-), M)$) to be invertible. The verdict is genuinely selective.

Example 8.4 (The effects sort themselves). The finite-distribution monad $\mathbf{D}$ is affine and codense, so $\llbracket - \rrbracket_{\mathbf{D}}$ is fully faithful (modulo the standard rigidity $\mathbf{Nat}(\mathbf{D}, \mathbf{D}) = \{\text{id}\}$). By contrast **Maybe**, exceptions $X + E$, the writer $E \times (-)$ and the reader $(-)^R$ ($|R| \geq 2$) all fail codensity and so lose fullness. The reader is affine but *not* codense ($\Phi(A) = (RA)^R \neq A^R = MA$), showing (i) and (ii) are independent conditions.

Remark 8.5 (The wrong criterion). The natural guess “ $\llbracket - \rrbracket_M$ is full iff M preserves coproducts” is false for the target $[\mathbf{Set}, \mathbf{Set}]$: that is the criterion for the coarser Kleisli-enriched target, and it gives the wrong verdict on $\mathbf{D}$. The writer $E \times (-)$ preserves coproducts yet fails codensity ($\Phi(1) = |E|^{|E|} \neq |E|$ for $|E| \geq 2$), a clean refutation. The controlling invariant is the codensity monad $\text{Ran}_M M$, not coproduct preservation.

The second invariant governs composition. Call M *polynomial* if $M \cong \sum_{i \in I} (-)^{B_i}$, i.e. $M = \llbracket [M] \rrbracket$ for a container $[M]$.

Theorem 8.6 (Composition = polynomiality; *peer-reviewed*). (Sufficiency, unconditional.) If M is polynomial then $p \triangleleft_M q := p \triangleleft [M] \triangleleft q$ satisfies $\llbracket p \triangleleft_M q \rrbracket_M \cong \llbracket p \rrbracket_M \circ \llbracket q \rrbracket_M$ naturally, so the essential image of $\llbracket - \rrbracket_M$ is closed under composition. The operation $\triangleleft_M$ is associative but has a two-sided unit up to container isomorphism only when $M = \text{Id}$: it is semigroupal, not monoidal. (Necessity, conditional.)

The converse — closure $\Rightarrow M$ polynomial — holds on an exactly delimited class strictly larger than the fully symmetric world: for super-polynomial M (a cardinality law, ruling out powerset, $\mathbf{D}$, ultrafilter, continuation); for analytic M with $M\emptyset = \emptyset$ (right-cancellation of plethysm, at unbounded degree and nesting); for analytic M of bounded wreath depth (an imprimitivity invariant subsuming multiset and symmetric powers); and for the free-algebra monad of every unital non-flat operad with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$. It fails for partially symmetric unital operads: a ternary operation with cyclic (C_3) slot-symmetry and an absorbed unit is non-polynomial yet composition-closed.

The one genuinely open point is the converse of the last clause ($\Delta_{C_2} \in \mathcal{S} \Rightarrow$ closed), verified for the all-binary inhabitant at small arities and open in general. Within the fully symmetric world — which contains all the applied effect monads (multiset, distribution, powerset) — composition $\iff$ flatness $\iff$ polynomiality.

Corollary 8.7 (The trade-off; peer-reviewed). *Fullness (codensity) and composability (polynomiality) are independent invariants of M that trade oppositely across the standard effects. The writer $E \times (-)$ is polynomial, hence composes, but is not codense, hence not full. The distribution monad $\mathbf{D}$ is affine and codense, hence full, but is not polynomial, hence does not compose. Probability keeps faithful naming and loses composition; error keeps composition and loses faithful naming. In particular the folklore “composes $\iff$ commutative and affine” is false: $\mathbf{D}$ is affine and commutative but does not compose, while **Maybe** is neither yet does.*

8.3 Positive suite II: the change-of-base census

Beyond the Kleisli identification, the same frame $\mathbf{Fam}((-)^{\text{op}})$ absorbs a census of monad/comonad transports, each a change of base or a fibrewise transport read in the appropriate variance. All are *proved* in the companion material.

Proposition 8.8 (Position transfer: monads become comonads; *proved*). *Because positions are contravariant, a monad M on **Set** postcomposed on the position family, $G(S, P) = (S, M \circ P)$, is a comonad on **Cont** — with counit η and comultiplication μ on the fibres — and this is a biconditional: the comonad laws for G hold iff the monad laws for M do (single-shape containers realise every fibre). Globally $G = \mathbf{Lan}_{(S, P)} M$. Dually a comonad on **Set** yields a monad on **Cont**. This is the slogan of the whole frame in miniature: a monad on the base is a comonad on the fibrewise opposite.*

Proposition 8.9 (The $\forall/\exists$ liftings; *proved*). *Quantifying a predicate container over positions gives two liftings. The $\exists$ -lifting is exactly composition, $E = \triangleleft$, a bifunctor on all of **Cont** with unit $y^{(-)}$. The $\forall$ -lifting $A = \Pi$ is a bifunctor only on cartesian morphisms in its first argument (the pushforward is indexed by set-sections of the backward position map, which are unique iff the map is a bijection), and on that domain it is a left module of $(\mathbf{Cont}, \triangleleft, y^{(-)})$. The mixed comparison $AX(EYC) \rightarrow E(AXY)C$ is an isomorphism iff X is linear. This is the semantic content of M lifting to containers iff M is cartesian, seen one level down.*

Proposition 8.10 (The free monad on a container; *proved*). *A signature is a container, and its free monad — the type of effect trees — is a monad on **Cont** built by iterated $\triangleleft$ -composition. In the linear instance over a monoidal base, the free $\triangleleft$ -monoid on the one-shape container $C = (\star, A)$ is $C^* = (\mathbb{N}, (A^{\otimes n})_n)$, with $\llbracket C^* \rrbracket(W) = \bigoplus_{n \geq 0} A^{\otimes n} \otimes W = T(A) \otimes W$ the tensor algebra: over **Vec** this is O’Neill’s free monad, realised functorially.*

The composition question of §8.2 is the fifth census entry, and the position transfer and liftings are the fibrewise ones; naming the frame turns a list of theorems into a single diagram.

8.4 Positive suite III: the four monoidal structures

The base **Cont** itself carries four monoidal structures, previewed in Chapter 3, §3.4: the coproduct $+$, the product $\times$, the Dirichlet tensor $\otimes$ (Definition 3.10; on containers the shapes and positions both multiply, cf. Proposition 3.11), and the substitution operator $\triangleleft$. Their full classification is a topic in its own right; we record here only the map and the closure results the rest of the dossier uses. All are *proved*; the Lean certifications are noted where they exist.

Why exactly these four. Every monoidal structure $(\star, I)$ on **Set** induces one on **Cont** by Day convolution [1, 31]: with $p \otimes_\star q = (S_p \times S_q, (s, t) \mapsto p_s \star q_t)$ and unit y^I . Call a structure *convolutional* if it preserves coproducts in each variable and closes the representables.

Theorem 8.11 (Day classification; *proved*). $(\star, I) \mapsto (\otimes_\star, y^I)$ is an equivalence from monoidal structures on **Set** to convolutional monoidal structures on **Cont**. Two of the four are its values, $\times = \otimes_{(+, \emptyset)}$ and $\otimes = \otimes_{(\times, 1)}$; the coproduct $+$ and the substitution $\triangleleft$ lie outside the family (they fail coproduct-preservation in a variable). This explains the “upside-down” arithmetic — the product’s positions are a disjoint union because it is the Day convolution of the coproduct of sets, $y^{A+B} \cong y^A \times y^B$.

Theorem 8.12 (Pointwise = product; *proved*). Among all monoidal structures on **Cont** — Day or not — the categorical product $\times$ is the unique one whose extension is the pointwise product of functors, $\llbracket p \ast q \rrbracket \cong \llbracket p \rrbracket \times \llbracket q \rrbracket$. Being pointwise is the rare property, not the generic one; in particular $\llbracket - \rrbracket$ is strong monoidal into $([\mathbf{Set}, \mathbf{Set}], \times)$ only for the cartesian structure.

Theorem 8.13 (The comparitor; *proved*). For each p , $p \otimes (-)$ is the left Kan extension of $p \triangleleft (-)$ along the representable embedding, and the comparison $p \otimes q \rightarrow p \triangleleft q$ is the counit of the resulting coreflection: $p \otimes (-)$ is the terminal coproduct-preserving approximation to $p \triangleleft (-)$. The Dirichlet tensor is the substitution operator with its dependency switched off — and not by fiat: it is the Day-ification of $\triangleleft$, which is why the two agree on representables ($y^A \otimes y^B = y^{A \times B} = y^A \triangleleft y^B$).

Closures. The Dirichlet tensor is closed, with internal hom the container whose *shapes* are the container morphisms [31]; this closure is machine-checked — $(\mathbf{Cont}, \otimes, y^{(-)})$ is a closed monoidal category in Lean 4, currying and uncurrying inverse on the nose (*lean-verified*, `Dirichlet.lean`). The substitution operator $\triangleleft$ is *not* left-closed but carries a right coclosure, whose comonoids are the directed containers, hence the small categories [26]. And **Cont** is cartesian closed but *not* locally cartesian closed, so it does not model dependent type theory — a fact worth stating plainly, and one the frontier chapter revisits.

Interaction. The pair $(\otimes, \triangleleft)$ is a normal duoidal category [33], and its interchange law houses the comparitor. Its double comonoids — objects that are comonoids for both tensors compatibly — are classified: they are exactly the *sets of commutative monoids*, by a fibrewise Eckmann–Hilton collapse (*proved*), correcting the plausible guess “the degenerate polynomials” in both directions.

8.5 Positive suite IV: when does substitution exist over a base?

The Dirichlet tensor $p \otimes q = (S \times T, (P_s \otimes Q_t))$ exists over *every* closed monoidal base. The substitution product $\triangleleft$ — the one whose comonoids are categories — is different: it mostly does not exist, and where it does, its existence is not a matter of degree but a sharp theorem. Call $\mathcal{C}$ $\triangleleft$ -*admissible* if the image of $\llbracket - \rrbracket$ is closed under composition, $\llbracket p \triangleleft q \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$ for all p, q . All results below are *proved* (the $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ result on an explicitly stated locus).

Everything is controlled by one necessary condition, obtained by evaluating any candidate composite at the terminal object $\mathbf{1} = [0, X]$.

Lemma 8.14 (Shape Lemma; *proved*). *If $\mathcal{C}$ is $\triangleleft$ -admissible then for every P and every set T the internal hom $[P, T \cdot \mathbf{1}]$ is a copower of $\mathbf{1}$ (here $T \cdot \mathbf{1} = \coprod_T \mathbf{1}$). The base must see the external shape set inside itself, as a copower of $\mathbf{1}$.*

On the set-like extreme admissibility does not merely coexist with the rest of the structure — it forces it.

Theorem 8.15 (Extensive rigidity and the forced package; *proved*). *Let $\mathcal{C}$ be nontrivial, infinitary leftextensive and cartesian closed. If $\mathcal{C}$ is $\triangleleft$ -admissible then its unit $\mathbf{1}$ is connected ($\mathcal{C}(\mathbf{1}, -)$ preserves small coproducts); consequently $\llbracket - \rrbracket$ is fully faithful and the reindexing functor $F_q = \text{Fam}(\llbracket q \rrbracket^{\text{op}})$ is left adjoint to $(-) \triangleleft q$ for every q . In particular $\mathbf{Set} \times \mathbf{Set}$ is inadmissible, its unit $(1, 1)$ being disconnected. Fullness of $\llbracket - \rrbracket$ and existence of the reindexing left adjoint are literally the same comparison lemma applied twice.*

The converse fails — connectedness alone is not enough — and the exact condition upgrades the necessary Shape Lemma to a sufficient one.

Theorem 8.16 (Characterisation; *proved*). *For $\mathcal{C}$ nontrivial, infinitary leftextensive and cartesian closed, $\mathcal{C}$ is $\triangleleft$ -admissible iff the Shape Lemma holds, equivalently iff an external distributive law holds. When $\mathcal{C}$ has a connected-components functor $\pi_0 \dashv (-) \cdot \mathbf{1}$ (as every Grothendieck topos does), this reads: $\mathcal{C}$ is $\triangleleft$ -admissible iff π_0 preserves finite products.*

This single criterion unifies the two ways admissibility can fail. The disconnected unit of $\mathbf{Set} \times \mathbf{Set}$ and the connected-unit obstruction below are one phenomenon, $\pi_0(X \times Y) \not\cong \pi_0 X \times \pi_0 Y$; it is co-extensive with the *stable-units* property of the connected-components reflection in categorical Galois theory [48], though established here via the distributive law rather than by descent.

Theorem 8.17 (Connected is not sufficient; *proved*). *The Artin gluing $\text{Gl}((-)^2)$ is a Grothendieck topos with a connected unit that is nonetheless $\triangleleft$ -inadmissible: its π_0 is not multiplicative, since for the single edge K one has $\pi_0(K \times K) = 2 \neq 1$ (Weichsel’s theorem [47]). This is exactly the gap between the internal (Gambino–Kock) composition, which every topos supports [32], and the external-shape composition of $\llbracket - \rrbracket$.*

On the opposite, additive pole the situation inverts. Over $\mathbf{Vec}_{\text{fd}}$ every object is copower-tiny, so $\triangleleft$ collapses onto the Dirichlet tensor, $p \triangleleft q = p \otimes q$, and $(-) \triangleleft q$ has a left adjoint only when q has a single shape; the unit is always disconnected. Between the poles the admissible region is thin but inhabited: $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ is admissible (on its locus) with a disconnected unit, combining a rigid distributive factor with a flexible copower-tiny one — so the extensive-pole characterisation does not extend to a global one (*proved*). The organising fact behind the whole map is determinacy.

Theorem 8.18 (Determinacy; *proved*). *If $\mathbf{1}$ is connected then $\llbracket - \rrbracket$ is injective on objects up to isomorphism, so $p \triangleleft q$ is determined by $\mathcal{C}$ and admissibility is a property of $\mathcal{C}$. If $\mathbf{1}$ is disconnected this can fail: over $\mathbf{Vec}_{\text{fd}}$ the objects $(\{*\}, k^2)$ and $(\{1, 2\}, k)$ present the same functor $X \mapsto X \oplus X$ but are not isomorphic, so $\triangleleft$ becomes a choice and the base-change question stops being about $\mathcal{C}$ at all.*

8.6 The linear wall: $\mathbf{Vec}$ is neither full nor faithful

The sharpest instrument for separating the four notions of “container in a category” — external families $\mathbf{Fam}(\mathcal{C}^{\text{op}})$, indexed dependent-polynomial functors over an LCC base [32], fibrational containers [43], and Walker’s locally subcartesian closed weakening [44] — is a base that fails *both* extensivity and local cartesian closure. The category $\mathbf{Vec}$ of vector spaces is exactly such a base: the comparison $\coprod^{\mathbf{Set}} \rightarrow \oplus$ is not an isomorphism, and there is no internal dependent product Π . Off the “good” zone a base fails for exactly one of two independent reasons, loss of extensivity (shapes no longer recoverable, $\llbracket - \rrbracket$ no longer full) or loss of internal Π ($\triangleleft$ and closedness collapse); toposes lose neither, quantales and affine bases lose only the second (reached by Walker’s affine repair), and $\mathbf{Vec}$ and $R\text{-Mod}$ lose both, so only the external approach reaches them. Three theorems make this quantitative; all are *proved*.

Theorem 8.19 (Fullness = unit-connectedness; *proved*). *For any closed symmetric monoidal $\mathcal{C}$ with small coproducts, $\llbracket - \rrbracket$ is fully faithful iff the monoidal unit I is connected ($\mathcal{C}(I, -)$ preserves small coproducts). This is not extensivity of $\mathcal{C}$: $\mathbf{Set} \times \mathbf{Set}$ is (l)extensive yet its unit $(1, 1)$ is disconnected and $\llbracket - \rrbracket$ is not full (the two “crossed” transformations are unrealised), while $\mathbf{Set}$ is full not because it is extensive but because 1 is connected. It recovers Abbott–Altenkirch–Ghani [24] by naming that hidden hypothesis. The reconstruction of a container from an abstract familiarly representable functor is Diers’ [42], and uses extensivity of the codomain, not the base — which is why the folklore said “extensivity”.*

Corollary 8.20 (The linear collapse; *proved*). *Over $\mathcal{C} = \mathbf{Vec}$ the unit is k and $\mathbf{Vec}(k, -)$ is the underlying-set functor, which does not preserve $\oplus$: the comparison $\coprod_t^{\mathbf{Set}} X_t \rightarrow \oplus_t X_t$ is neither surjective (it misses cross-summand sums) nor injective (it collapses the zero vector, which sits in every summand). Hence over $\mathbf{Vec}$ the extension $\llbracket - \rrbracket$ is neither full nor faithful — the failure of faithfulness being precisely that $\oplus$ collapses tagged zeros. This is the honest strength of the linear denotation: the on-the-nose container/functor correspondence that holds over $\mathbf{Set}$ breaks in both directions over an additive base.*

The other two theorems measure the remaining obstructions, on independent seams. The Dirichlet tensor $\otimes$ is closed iff a familial-representability functor Φ is representable; over the linear base this holds *only* on the corner $\mathbf{Fam}_{\text{fin}}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$ of finite shape-sets and finite-dimensional positions (the load-bearing conjunct is *dualizable-and-summable*), failing over both $\mathbf{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$ and $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ by dual mechanisms (*proved*). The substitution product $\triangleleft$ is left-closed exactly where it collapses onto $\otimes$ — over a base whose positions are dualizable (tiny), $\triangleleft = \otimes$ and the left hom is the Dirichlet hom — so it is left-closed on $\mathbf{Fam}_{\text{fin}}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$ and obstructed over every extensive base (*proved*). Extensivity is thus *opposed* to $\triangleleft$ -left-closedness: the property whose failure costs fullness and Dirichlet closedness is exactly what would buy substitution closedness. These are the seams Weber’s distributivity- δ and familial-functor theories [45, 46] place on different rungs; the composition-monoid refinement of the indexed corner is [50], and the general-base tensor and its comonoids-as-enriched-categories are Dorta–Jarvis–Niu [49].

8.7 The $\{\text{Id}\}$ collapse

The linear wall shows a *specific* base failing. The final result is stronger and structural: over *no* nontrivial effect does the natural self-enriched semantics enjoy a uniform full-and-faithfulness theorem. The apparent counterexample — the reader monad, whose Kleisli category is cartesian closed — is the pivot that makes the point.

There are three extensions of an M -container into endofunctors, with three notions of naturality. Fix a monad M and write $\mathcal{C} = \text{Kl}(M)$.

- (A) the *ordinary* extension $\llbracket - \rrbracket_M : \text{Fam}(\text{Kl}(M)^{\text{op}}) \rightarrow [\mathbf{Set}, \mathbf{Set}]$, whose morphisms are pure natural transformations: faithful always, full iff M codense and each $\mathbf{Set}(A, M-)$ connected (Theorem 8.3);
- (B) the *Kleisli-presheaf* extension $\llbracket - \rrbracket^{\text{Kl}} : \text{Fam}(\text{Kl}(M)^{\text{op}}) \rightarrow [\text{Kl}(M), \mathbf{Set}]$, a coproduct of representables: by ordinary co-Yoneda it is *always* full and faithful, at the price that its morphisms are Kleisli-natural;
- (C) the *self-enriched* extension, defined when $\mathcal{C}$ is monoidal closed, $\llbracket S, P \rrbracket = \sum_s [P_s, -] : \mathcal{C} \rightarrow \mathcal{C}$, which realises the enriched-Yoneda desideratum $\text{Nat}(\sum_s [P_s, -], F) \cong \prod_s F(P_s)$ [39].

The three are one factorisation, $\llbracket - \rrbracket_M = R \circ \llbracket - \rrbracket^{\text{Kl}}$ with $R = (-) \circ F$ restriction along the free functor $F : \mathbf{Set} \rightarrow \text{Kl}(M)$; all loss of ordinary fullness lives in R , and its size is exactly the codensity monad $\text{Ran}_M M$.

Two independent invariants govern (C):

$$\mathcal{CL} = \{M : \text{Kl}(M) \text{ monoidal closed}\}, \quad \mathcal{UN} = \{M : \text{the monoidal unit of } \text{Kl}(M) \text{ is connected}\}.$$

Closedness ($\mathcal{CL}$, available by Kock's criterion [40, 41]) makes (C) *exist*; connectedness ($\mathcal{UN}$, i.e. M preserves small coproducts, since $\text{Kl}(M)(1, X) = MX$) makes it *full* (Theorem 8.19). They are logically independent — the cartesian-closed reader sits in $\mathcal{CL} \setminus \mathcal{UN}$, the writer in $\mathcal{UN} \setminus \mathcal{CL}$ — and over a Kleisli base they meet in exactly one monad.

Lemma 8.21 (Closed + initial $\Rightarrow$ terminal; *lean-verified*). *In any symmetric monoidal closed category with an initial object $\emptyset$, the object $[\emptyset, X]$ is terminal for any X . (Each $(-) \otimes A$ is a left adjoint, so $C \otimes \emptyset \cong \emptyset$, whence $\mathcal{K}(C, [\emptyset, X]) \cong \mathcal{K}(\emptyset, X) \cong 1$.) The argument uses only monoidal closedness and an initial object — neither cartesian closedness nor connectedness, the two independent axes.*

Theorem 8.22 ($\mathcal{CL} \cap \mathcal{UN} = \{\text{Id}\}$; *proved, core lean-verified*). *Let M be a commutative strong monad on $\mathbf{Set}$. The self-enriched M -container extension $\sum_s [P_s, -] : \text{Kl}(M) \rightarrow \text{Kl}(M)$ is full and faithful if and only if $M \cong \text{Id}$. Equivalently $\mathcal{CL} \cap \mathcal{UN} = \{\text{Id}\}$: no nontrivial effect — not even an idempotent (localization) monad — carries a uniform enriched full-and-faithful semantics.*

Proof sketch. Fullness of (C) $\iff M \in \mathcal{CL} \cap \mathcal{UN}$. *Step 1:* $\mathcal{UN}$ forces the writer form. If M preserves small coproducts then the copower-comparison $T \times M1 \rightarrow MT$ is a natural bijection, so $M \cong (-) \times E$ with $E = M1$, and the monad laws are the laws of a (commutative) monoid on E . *Step 2:* $\mathcal{CL}$ forces $E \cong 1$. The base $\text{Kl}(M)$ is monoidal closed with initial object $\emptyset$ (since $\text{Kl}(M)(\emptyset, B) = 1$), so Lemma 8.21 supplies a terminal $\top$; then $\mathbf{Set}(C, M\top) = \text{Kl}(M)(C, \top) \cong 1$ forces $M\top \cong 1$, and the writer form gives $|\top| \cdot |E| = 1$, so $E \cong 1$ and $M \cong \text{Id}$. *Step 3:* Id qualifies ($\mathbf{Set}$ cartesian closed, $\mathbf{Set}(1, -) = \text{Id}$ preserves coproducts). $\square$

The core of this collapse — the closed-plus-initial-forces-terminal engine of Lemma 8.21, together with the resulting identity-only classification — is machine-checked in Lean 4 (*lean-verified*, `EnrichedFFCCollapse`).

Corollary 8.23 (No idempotent escape; *proved*). *No nontrivial idempotent monad lies in $\mathcal{CL} \cap \mathcal{UN}$: membership in $\mathcal{UN}$ forces the writer form, and a writer monad is idempotent only for $E \cong 1$; the terminal monad $LX = 1$ already fails connectedness, $L(1 + 1) = 1 \neq 2 = L1 + L1$.*

Remark 8.24 (The reader, reconciled). The collapse dissolves the apparent paradox. Cartesian closedness of $\mathbf{Kl}(\mathbf{Reader}_S)$ buys the *availability* of (C), not its fullness: the unit 1 has global-points functor $(-)^S$, which fails to preserve coproducts for $|S| \geq 2$, so (C) is not full. The reader’s failure is enrichment-invariant — non-codensity in (A), a disconnected unit in (C), one obstruction wearing two masks — and (B) is full only for Kleisli-natural, not pure, morphisms. Enrichment relocates the obstruction; it does not remove it.

The boundary, and the handoff. The chapter has traced how far the correctness equation generalises. It survives change of base wholesale — as *m*-containers with two computable invariants (§8.2), as the census of transports (§8.3), as the four monoidal structures with their closures (§8.4), and as substitution over exactly the π_0 -multiplicative bases (§8.5) — but it does *not* survive as a *uniform self-enriched full-and-faithful* semantics over any nontrivial effect: that is the $\{\text{Id}\}$ wall of Theorem 8.22, a statement about general effect monads. A different wall stands over a *single* additive base: the linear denotation over **Vec** (§8.6) is already neither full nor faithful, and the Vec-admissibility question of whether $F = \text{Hom}(E, \bigoplus_{\mathbb{N}} -)$ is free pushes the boundary into set theory. That set-theoretic frontier — where the verdict becomes independent of ZFC — is the subject of the next chapter, Chapter 9.

Chapter 9

The set-theoretic frontier: Vec-admissibility and the Gap- S arc

This is the frontier of the Theory pillar, and it is open. Every obstruction so far in this document — the Zappa–Szép cocycle $[\omega] \in H^2$, the prunability descent class, the direction-functor cohomology — has been a concrete, *absolute* invariant: computed once, its verdict holds in every model of set theory. The admissibility of the linear base **Vec** is the first node of the programme that is not like this. Whether **Vec** is $\triangleleft$ -admissible turns out to be governed by the cardinal arithmetic of the continuum, and its exact truth value changes from one model of ZFC to another.

The result of this chapter is a single, sharp identity. The composability of the linear base reduces to the projectivity of one module M over the full linear ring $S = \text{End}_k(E)$, and the *minimal number of generators* of that module is exactly the dominating number:

$$\boxed{\mu_S(M) = \mathfrak{d}}.$$

A cardinal characteristic of the continuum is the load-bearing invariant of a functor-category admissibility question. From this identity the conjecture is *false* in every model with $\mathfrak{d} = \mathfrak{c}$ — in particular under CH, under Martin’s Axiom, and in the Cohen model — and the only surviving positive case is squeezed into the narrow window $2^{\aleph_0} = 2^{\mathfrak{d}}$, whose canonical witness is the random-real model. Whether M is projective there is the programme’s deepest open problem. The chapter ends the Theory pillar on that live edge, as it should.

Everything in §§9.1–9.3 is *proved* in ZFC; the residual of §9.5 is *open*. Nothing in this chapter is lean-verified, and nothing is externally published (Honest Ledger, items 1 and 4).

9.1 From admissibility to a module

Recall the denotation of a family $(S, P) \in \text{Fam}(\mathcal{C}^{\text{op}})$, the object that presents a container over the base $\mathcal{C}$:

$$\llbracket S, P \rrbracket X = \coprod_{s \in S} \mathcal{C}(P(s), X).$$

The base $\mathcal{C}$ is *$\triangleleft$ -admissible* when the image of $\llbracket - \rrbracket$ is closed under composition of endofunctors — the minimal condition under which the composition $p \triangleleft q$ of two directed containers has any denotational meaning over $\mathcal{C}$ (Chapter 9 uses only this weakest form; a non-existence result proved against it is maximally strong). Over **Set** every base is admissible; over **Vec** the unit object is 0 and the shape set of a composite becomes invisible, and the whole difficulty concentrates in a single instance.

Conjecture 9.1 (6.2 / Gap- S). *Over a countable field k , the additive endofunctor $F = \prod_{r \in \mathbb{N}} (\bigoplus_{m \in \mathbb{N}} (-))$ of $\mathbf{Vec}_k$ is a coproduct of representables; equivalently, $\mathbf{Vec}_k$ is $\triangleleft$ -admissible at the instance presenting $\bigoplus_{\mathbb{N}} (-)$.*

The reduction to algebra is the spine of the arc and is proved in the module note [8]. Let $E = k^{(\mathbb{N})}$ and $S = \text{End}_k(E)$, the ring of column-finite $\mathbb{N} \times \mathbb{N}$ matrices over k ; for countable k , $|S| = \mathfrak{c}$. Restriction to the subcategory of countable-dimensional spaces identifies additive endofunctors with left S -modules via $T \mapsto T(E)$, sending $\text{Nat}(T, T') \cong \text{Hom}_S(T(E), T'(E))$ and Ext^1 to Ext^1 . Under this dictionary the identity functor maps to the simple projective E , the countable coproduct $\bigoplus_{\mathbb{N}} (-)$ to $P = \bigoplus_{\mathbb{N}} E$ (a countably generated, non-finitely-generated projective), and the functor of the conjecture to

$$M = \prod_{r \in \mathbb{N}} P = \prod_{r \in \mathbb{N}} \bigoplus_{m \in \mathbb{N}} E.$$

Theorem 9.2 (Module reduction [8, Thm. 1–2]; proved, ZFC). *$\mathbf{Vec}_k$ is $\triangleleft$ -admissible at the instance of Conjecture 9.1 if and only if M is projective as a left S -module.*

Two structural facts about S drive everything below, and both are peculiar to the *noncommutative* full linear ring. First, S is von Neumann regular, so every module is flat and every submodule is pure. Second — and this is the identity that makes the arc subtle rather than routine —

$$\prod_{r \in \mathbb{N}} E \cong S \quad \text{as left } S\text{-modules}, \quad (*)$$

free of rank one, because a column-finite matrix *is* an arbitrary sequence of its columns [8, Thm. 2]. The identity $(*)$ is the homogeneity of S : its minimal idempotents are mutually conjugate, so a countable product of copies of the regular module reassembles into S . A commutative ring has no such homogeneity, and this is exactly why the tempting commutative surrogate k^ω is not faithful to the problem [12]; the phenomenon is intrinsically noncommutative.

Identity $(*)$ has an immediate and destructive consequence for the obvious attack. One would like to prove M non-projective by a Baer–Specker “product-is-bigger-than-coproduct” argument, exhibiting an exotic map that cannot lift. But $(*)$ makes the relevant product *free*, so the dual of M stays large ($\dim_k \text{Hom}_S(M, E) = \mathfrak{c}$) and no size-gap argument can succeed: the Baer–Specker mechanism is neutralised [8, Thm. 4]. Any refutation must count something finer than cardinality, and any positive proof must survive the collapse of the naive obstruction. This is the shape of the whole difficulty.

Before counting, one reduction removes the shortest independence route. Call a module *R-projective* (Dual Baer) if every map into a cyclic quotient S/I lifts through $S \twoheadrightarrow S/I$; projective always implies *R-projective*.

Theorem 9.3 (Dual-Baer collapse [9]; proved, ZFC). *M is projective if and only if M is *R-projective*.*

The proof uses that S contains a free left ideal of rank $\mathfrak{c}$ (an almost-disjoint-family construction inside $(*)$), which upgrades *R-projectivity* to projectivity for any module generated by at most $\mathfrak{c}$ elements; M is one, since $|M| \leq \mathfrak{c}$. Theorem 9.3 matters because it forecloses Trlifaj’s packaged independence theorem for the Dual Baer Criterion, which requires the ring to be *small semiartinian* — a hypothesis S fails. Whatever the status of Conjecture 9.1, it is not an off-the-shelf instance of known Dual-Baer independence.

9.2 The first count: false under CH

The original refutation counts the dual of a hypothetical decomposition. It rests on two ZFC facts about the fixed module M : it is *not* countably generated [8, §5] (a Bergman-style diagonalisation, invisible to cardinality since $|M| = \mathfrak{c}$), and its dual is small, $\dim_k \text{Hom}_S(M, E) = 2^{\aleph_0}$ [8, Thm. F].

Theorem 9.4 (First count [10]; proved, ZFC). *If M is projective then $2^{\aleph_0} = 2^{\aleph_1}$. Consequently, if $2^{\aleph_0} < 2^{\aleph_1}$ — in particular under CH and under GCH — then M is not projective, **Vec** is $\triangleleft$ -inadmissible, and Conjecture 9.1 is false.*

Proof sketch. By Kaplansky's theorem a projective module is a direct sum of countably generated submodules, $M = \bigoplus_{i \in I} M_i$ with each $M_i \neq 0$. If $|I| \leq \aleph_0$ then M is countably generated, a contradiction; so $|I| \geq \aleph_1$. Each nonzero countably generated projective has nonzero dual, so $\text{Hom}_S(M, E) = \prod_{i \in I} \text{Hom}_S(M_i, E)$ contains a copy of k^I , giving $\dim_k \text{Hom}_S(M, E) \geq 2^{|I|} \geq 2^{\aleph_1}$ by Erdős–Kaplansky. But the dual has dimension $2^{\aleph_0}$, so $2^{\aleph_1} \leq 2^{\aleph_0}$, forcing equality. $\square$

The parallel with the Baer–Specker group $\mathbb{Z}^{\mathbb{N}}$ is exact and instructive. There the same schema gives an *absolute* contradiction ($\mathbb{Z}^{\mathbb{N}}$ is non-free in ZFC), because Specker's theorem collapses the dual to countable rank. Over S the freeness $(*)$ keeps the dual large, and the contradiction degrades from “ $\aleph_0$ versus $2^{\mathfrak{c}}$ ” to merely “ $2^{\aleph_0}$ versus $2^{\aleph_1}$.” The void Baer–Specker is precisely what demotes an absolute refutation to a CH-conditional one: same count, different dual size, different verdict.

Theorem 9.4 is silent when $2^{\aleph_0} = 2^{\aleph_1}$. The original note nominated $\text{MA} + 2^{\aleph_0} = \aleph_2$ as the natural residual model in which independence might live. The next section removes exactly that model.

9.3 The sharp invariant: $\mu_S(M) = \mathfrak{d}$

The first count sees only the crude cut $|I| \geq \aleph_1$: the index set of a Kaplansky decomposition is uncountable. The sharpening computes its true size. Write $\mathfrak{d}$ for the dominating number — the least size of a family in $(\mathbb{N}^{\mathbb{N}}, \leq^*)$ cofinal for eventual domination, equivalently the least number of compact sets covering Baire space $\mathbb{N}^{\mathbb{N}}$ (a set is compact iff it is contained in some $\prod_r [0, g(r)]$). In ZFC, $\aleph_1 \leq \mathfrak{b} \leq \mathfrak{d} \leq \mathfrak{c}$.

For $x \in M$ write $\text{supp}(x) = \{(r, m) : x_{r,m} \neq 0\}$; each fibre $\text{supp}(x)_r = \{m : (r, m) \in \text{supp}(x)\}$ is finite (this is the block-column constraint defining $M \subseteq \prod_{r,m} E$). The action only shrinks supports: $\text{supp}(A \cdot x) \subseteq \text{supp}(x)$ for $A \in S$, because $(A \cdot x)_{r,m} = A(x_{r,m})$ vanishes wherever $x_{r,m}$ does.

Theorem 9.5 (The generator number is $\mathfrak{d}$; proved, ZFC). $\mu_S(M) = \mathfrak{d}$.

Proof. Lower bound $\mu_S(M) \geq \mathfrak{d}$. Let G generate M ; put $\kappa = |G|$, so $\kappa \geq \aleph_0$ and $|[G]^{<\omega}| = \kappa$. Fix $0 \neq v \in E$, and for each $f \in \mathbb{N}^{\mathbb{N}}$ let $x^f \in M$ be the *graph element* $x^f_{r,f(r)} = v$, all other entries 0; it lies in M (one nonzero entry per row) and has $\text{supp}(x^f) = \{(r, f(r)) : r\}$. Since G generates, $x^f = \sum_{g \in F} A_g g$ for a finite $F \subseteq G$ and $A_g \in S$; by support-shrinking and additivity, $\text{supp}(x^f) \subseteq \bigcup_{g \in F} \text{supp}(g)$. Setting $U_r^F = \bigcup_{g \in F} \text{supp}(g)_r$ (finite, a finite union of finite sets), this says $f(r) \in U_r^F$ for every r , i.e. $f \in \prod_r U_r^F$. Hence

$$\mathbb{N}^{\mathbb{N}} = \bigcup_{F \in [G]^{<\omega}} \prod_r U_r^F,$$

a covering of Baire space by $\leq \kappa$ compact sets. The least such number is $\mathfrak{d}$, so $\kappa \geq \mathfrak{d}$.

Upper bound $\mu_S(M) \leq \mathfrak{d}$. Fix a $\leq^*$ -dominating family $\{h_\xi : \xi < \mathfrak{d}\} \subseteq \mathbb{N}^\mathbb{N}$. For each ξ set $M_\xi = \{x \in M : x_{r,m} = 0 \text{ for } m > h_\xi(r)\}$. As a set $M_\xi = \prod_r E^{h_\xi(r)+1}$; reindexing by the countably infinite set $\{(r, m) : m \leq h_\xi(r)\}$ this is $\prod_{\mathbb{N}} E \cong S$ by $(*)$, so M_ξ is *cyclic*, $M_\xi = S \cdot g_\xi$. Given $x \in M$, let $N_x(r) = \max \text{supp}(x)_r$; choose ξ with $N_x \leq^* h_\xi$, say $N_x(r) \leq h_\xi(r)$ for $r \geq r_0$. Then $x = x \cdot \mathbf{1}_{r \geq r_0} + x \cdot \mathbf{1}_{r < r_0}$: the first summand lies in M_ξ , the second has finite total support, hence lies in the countably generated $M_{\text{fin}} = \bigoplus_{r,m} E$. Thus $\{g_\xi : \xi < \mathfrak{d}\}$ together with countably many matrix units generate M , and $\mu_S(M) \leq \mathfrak{d} + \aleph_0 = \mathfrak{d}$. $\square$

The proof turns on the two faces of $\mathfrak{d}$: the lower bound is $\mathfrak{d} = \text{cov}(\text{compact})$, the upper bound is $\mathfrak{d} = \min |\text{dominating family}|$. The upper bound uses $(*)$ in an essential way — the same freeness that voids Baer–Specker is what collapses each “staircase region” M_ξ to a *cyclic* module, so that the only cost of generating M is the growth of supports (which is $\mathfrak{d}$), never the values.

9.4 The sharpened refutation, and why it matters

The generator number feeds the dual count directly, replacing the crude $|I| \geq \aleph_1$ by $|I| \geq \mathfrak{d}$.

Theorem 9.6 (Sharpened count; proved, ZFC). *If M is projective then $2^\mathfrak{d} = 2^{\aleph_0}$. Consequently, if $2^{\aleph_0} < 2^\mathfrak{d}$ then M is not projective and Conjecture 9.1 is false.*

Proof. A Kaplansky decomposition $M = \bigoplus_{i \in I} M_i$ has $|I| \geq \mu_S(M) = \mathfrak{d}$ by Theorem 9.5 (the summands’ generators generate M). As in Theorem 9.4, $\dim_k \text{Hom}_S(M, E) \geq 2^{|I|} \geq 2^\mathfrak{d}$, while $\dim_k \text{Hom}_S(M, E) = 2^{\aleph_0}$; so $2^\mathfrak{d} \leq 2^{\aleph_0}$, and equality follows since $\mathfrak{d} \geq \aleph_0$. $\square$

Corollary 9.7 (The false region; proved, ZFC). *If $\mathfrak{d} = \mathfrak{c}$ then Conjecture 9.1 is false. In particular it is false under CH, under Martin’s Axiom $\text{MA} + \neg\text{CH}$, and in the Cohen model.*

Proof. If $\mathfrak{d} = \mathfrak{c}$ then $2^\mathfrak{d} = 2^\mathfrak{c} > 2^{\aleph_0}$ by Cantor, so Theorem 9.6 applies. CH gives $\mathfrak{d} = \aleph_1 = \mathfrak{c}$; MA gives $\mathfrak{d} = \mathfrak{c}$; and adding Cohen reals makes $\text{cov}(\mathcal{M}) = \mathfrak{c} \leq \mathfrak{d}$, so $\mathfrak{d} = \mathfrak{c}$ there as well [6, 5]. $\square$

This strictly enlarges the false region of the first count, and it does so by eating the very model the first count had to leave open. Under $\text{MA} + 2^{\aleph_0} = \aleph_2$ one has $2^{\aleph_0} = 2^{\aleph_1}$, so Theorem 9.4 is silent — this was the nominated residual model for possible independence. But MA forces $\mathfrak{d} = \mathfrak{c}$, so Corollary 9.7 *refutes* the conjecture there. The sharpening removes $\text{MA} + \neg\text{CH}$ from the residual entirely.

The crown-jewel reading. Theorem 9.5 is the reason this arc is worth its length. A question about the composition of directed containers over the linear base — whether two containers substitute into a third — has been reduced to a single cardinal, and that cardinal is not $\aleph_1$ or $\mathfrak{c}$ but the dominating number $\mathfrak{d}$, a genuine characteristic from the Cichoń diagram. The composability boundary of a categorical semantics is thus located inside the combinatorics of Baire space. To the author’s knowledge this is the first place in the container programme — and one of very few places anywhere in the semantics of computation — where a definability/composability question is answered by a cardinal characteristic of the continuum rather than by a fixed invariant.

Varying the ground division ring cannot force the issue. One might hope to make the refutation *absolute* by tuning a parameter: replace k by a division ring D , so $S_D = \text{End}_D(V)$ with $V = D^{(\mathbb{N})}$. The freeness $(*)$ holds over every D by the same matrix-unit homogeneity, so the generator number is unchanged — $\mu_{S_D}(M) = \mathfrak{d}$ for every D — while only the dual ceiling moves:

M projective now forces $|D|^{\mathfrak{d}} = |D|^{\aleph_0}$. The lower bound stays pinned at $\mathfrak{d}$; only one side of the coincidence slides with $|D|$.

Corollary 9.8 (Easton silence; proved, ZFC). *For every division ring D it is consistent with ZFC that $|D|^{\mathfrak{d}} = |D|^{\aleph_0}$; hence no choice of D makes the counting obstruction prove M non-projective in ZFC.*

For $|D| \leq \mathfrak{c}$ this is the random-real model ($\mathfrak{d} = \aleph_1$, $2^{\aleph_0} = 2^{\aleph_1}$); for larger $|D|$ with $\text{cf}(|D|) > \mathfrak{d}$ it follows from Easton's theorem on the freedom of the continuum function. The honest scope matters: Corollary 9.8 shows the *lever* cannot force falsity in ZFC; it does *not* construct M projective. Varying D relocates the computation but does not decide it.

9.5 The residual, honestly

The two counts together confine any surviving positive case to the window

$$2^{\aleph_0} = 2^{\mathfrak{d}}, \quad \text{equivalently} \quad \mathfrak{d} < \mathfrak{c},$$

strictly smaller than the first count's $2^{\aleph_0} = 2^{\aleph_1}$. The canonical model is the *random-real model*, where $\mathfrak{b} = \mathfrak{d} = \aleph_1 < \mathfrak{c} = \aleph_2$ and $2^{\aleph_0} = 2^{\aleph_1} = \aleph_2$; there $\mu_S(M) = \mathfrak{d} = \aleph_1$ and the count permits a decomposition $M = \bigoplus_{\aleph_1} (\text{countably generated projective})$. Whether that decomposition exists is the open crux.

Why the naive positive route fails is itself diagnostic, and worth stating precisely, because it locates where any future proof must act. One is tempted to build the decomposition from the staircase filtration $M = \bigcup_{\xi < \aleph_1} M_\xi$ of Theorem 9.5: each $M_\xi \cong S$ is projective, and each successor quotient $M_{h_{\xi+1}}/M_{h_\xi} \cong \prod_r E^{h_{\xi+1}(r)-h_\xi(r)}$ is again a countable product of E 's, hence projective. By Eklof's lemma a *continuous* well-ordered filtration with projective quotients would force M projective. The route fails at a single point: the filtration is *not continuous at limits*. For a limit λ of cofinality ω , $\bigcup_{\xi < \lambda} M_{h_\xi} \subsetneq M_{\sup_\xi h_\xi}$ — an element whose support grows to $\sup_\xi h_\xi$ pointwise but is dominated by no single h_ξ is excluded — and repairing the gap by taking unions destroys the clean $\prod E$ form of the successor quotient. The non-projectivity, if present, is stored in the post-limit quotients. This is exactly the Whitehead/ Γ -invariant signature: the obstruction lives at the limit stages of an $\aleph_1$ -filtration, not in the generator count. $\mu_S(M) = \mathfrak{d}$ supplies the *length* of the filtration but not a projective *continuous* one.

Question 9.9 (the residual; open). Is M projective in the random-real model — equivalently, can $\text{pd}_S M = 0$ hold in a model of $2^{\aleph_0} = 2^{\mathfrak{d}}$? A ZFC proof that $\text{pd}_S M \geq 1$ in every model makes Conjecture 9.1 false in ZFC; a decomposition in the random-real model makes it independent of ZFC.

The reframing through projective dimension [11] makes the question recognisable: the cokernel of M inside the free module $\Phi \cong S$ is $\prod_{\mathbb{N}} N_0$, a countable product of copies of a fixed projective-dimension-one seed $N_0 = S/I \cong \prod_{\mathbb{N}} E / \bigoplus_{\mathbb{N}} E$, so Conjecture 9.1 is exactly the Osofsky–Jensen question of whether a product raises projective dimension, sharpened to name the continuum. The ZFC bracket $1 \leq \text{pd}_S M \leq d+1$ (with $d = \min\{n : 2^{\aleph_0} \leq \aleph_n\}$) holds unconditionally.

The evidence is genuinely two-sided, and it is worth stating at honest weight. *Toward independence*: the operative threshold is the collapse $2^{\aleph_0} = 2^{\mathfrak{d}}$, an equalisation of the continuum function that forcing can arrange while preserving reflection, and $\text{pd}_S M = 0$ is a Whitehead-shaped statement about an $\aleph_1$ -indexed decomposition — the archetype of an independent assertion; the large dual (from $(*)$) removes the absolute obstruction that makes $\mathbb{Z}^{\mathbb{N}}$'s non-freeness a theorem of ZFC.

Toward false-in-ZFC: M is a canonical continuum-sized product, the exact shape of Osofsky’s high-projective-dimension witnesses; and the failure of “flat Mittag-Leffler $\Rightarrow$ projective” over S is already an ω_1 -scale phenomenon witnessed in ZFC [7], which the residual window constrains the continuum function without obviously demoting. The structural evidence leans toward independence, but this is an evidential lean, not a proof, and false-in-ZFC remains a live competitor. Deciding between them is Question 9.9, and it must be settled *natively* over S : the commutative surrogate k^ω is not faithful, precisely because the freeness $(*)$ that carries the whole reduction is intrinsically noncommutative [12].

This is where the Theory pillar ends: not on a solved theorem but on a sharp open one, with the boundary of the linear base’s composability pinned to a cardinal characteristic of the continuum and the last question — projectivity in the random-real model — honestly unanswered.

Part II

Applications

Chapter 10

GA / island-model composition

Genetic-algorithm island models compose subpopulations along a migration topology. That topology *is* the distributive law; whether the composite evolutionary system is coherent is whether the Zappa–Szép product exists. Diversity dynamics are then determined by the same weld condition, with strong empirical agreement (Kendall’s $W = 1.0$ across the measured runs).

[Assembly note: computed; migration topology = the distributive law.]

Chapter 11

Blockchain: re-entrancy as a failed distributive law

Smart contracts are coalgebras for $p(X) = (O \times X)^I$; composing two contracts requires a distributive law between them. Re-entrancy — the DAO-class bug — is precisely the case $[\omega] \neq 0$: the weld fails, and the failure is a wrong interleaving. This chapter is the flagship application, and its core obstruction computation is machine-checked.

[Assembly note: peer-reviewed; the DAO-class bug is $[\omega] \neq 0$.]

Chapter 12

Tax computation (Catala)

Tax law composes: an ontology DAG of definitions, scope composition of rules, and comonadic event-sourced projections of taxable events. The categorical picture is clean and suggestive. *Honest gap*: there is no formal artifact — no proof, no Lean, no running Catala integration. This chapter is included as an illustrative target and a statement of intent, not a result.

[Assembly note: NO artifact (Honest Ledger item 2). Frame as illustrative-only.]

Chapter 13

Agent orchestration: the harness is a worker

An agent harness is a worker in the store-comonad sense: it carries state and observes it. Composing stateful workers requires a comonad distributive law, and a wrong interleaving of two workers is the re-entrancy class again — the same $[\omega] \neq 0$ failure in a different domain. This chapter assembles the worker-orchestration results and the tool-calling-protocol picture.

[Assembly note: orchestration-ZS, workers-type-hierarchy proved.]

Chapter 14

Security audits as failed distributive laws

State-divergence bugs of the class surfaced by professional audits (Trail-of-Bits style) are, structurally, failed distributive laws: two operations that do not commute past each other in the way the intended composite assumes. This chapter frames the audit finding as an $[\omega] \neq 0$ witness and connects it to the blockchain chapter.

[Assembly note: STUB; extends the blockchain obstruction picture to audit findings.]

Chapter 15

Supply chains

Supply-chain composition welds supplier and consumer processes; the existence of a coherent global schedule is a Zappa–Szép condition. This chapter is a stub built from the supply-chain-ZS registry entry.

[Assembly note: STUB; computed in registry, needs a written body.]

Part III

Formalisation

Chapter 16

Milestones M1–M6: status

The equivalence chain is tracked as a sequence of formalisation milestones. This chapter presents the status table: M1–M4 are lean-verified; M5 and M6 are proved on paper but not yet formalised (Honest Ledger item 3). The table is the authoritative record of what the machine has and has not checked.

[Assembly note: build the M1–M6 status table from the registry JSON; M5–M6 rows are open.]

Chapter 17

The sorry-free Lean corpus

The programme maintains a sorry-free Lean 4 corpus of 66 modules under **Containers/**. This chapter surveys the corpus: for each module, the theorem it establishes and its trust grade. Building the full module $\rightarrow$ theorem $\rightarrow$ grade table is outstanding work.

[Assembly note: lean-verified (66 modules); per-module theorem/grade table to be built (GAP).]

Chapter 18

H^2 -engines and the free/cofree comonad in Lean

The obstruction machinery of Part I is partly machine-checked: the free and cofree comonad constructions with their universal properties, and the cyclic and fibre H^2 “engines” that compute the holonomy classes. This chapter assembles those Lean developments.

[Assembly note: lean-verified; free/cofree UP + H^2 engines.]

Chapter 19

Change-of-base adjunction and collapse theorems in Lean

The change-of-base story of §I.5 has a machine-checked core: the composition left adjoint $F_q \dashv (-) \triangleleft q$ with naturality, and the collapse theorems — the **Vec** biproduct collapse, the QPF abs/repr collapse, and the enriched full-and-faithful $\{\text{Id}\}$ collapse. This chapter assembles them.

[Assembly note: lean-verified; adjunction + three collapse theorems.]

Part IV

Impact

Chapter 20

Where compositional correctness has economic consequences

The applications are not chosen at random: they are the domains where a wrong weld costs money. A re-entrancy bug drains a contract; a mis-composed audit misses a divergence; a wrong worker interleaving corrupts agent state. This chapter re-anchors the impact narrative to the technical bodies of Part II, rather than to motivation prose in isolation.

[Assembly note: re-anchor motivation to the Part II technical bodies.]

Chapter 21

The AI + Lean pipeline: honesty and trust-grade methodology

This is the programme’s genuine differentiator, and it is a *method*, not a boast. Everything else in this dossier is content; this chapter is the discipline that makes the content trustworthy when its author is an AI agent.

21.1 The problem this solves

The premise is now demonstrated in public. AI-authored Lean proofs are real: a 2026 refutation of Köthe’s conjecture — a ninety-year-old open problem — ships with a Lean 4 formalisation authored autonomously by a pre-release system and then human-reviewed [20]. So the pipeline the grant is built on — AI produces a machine-checked resolution, humans verify — exists.

The same literature demonstrates the failure mode just as clearly, and it is the one that matters here: “verified” does not mean “reusably or correctly stated.” Ilin and Nugent autoformalised Grothendieck vanishing to a *zero-sorry* proof on the first pass, yet expert review found the definitions, the theorem’s generality, and the API unusable as a library contribution [21]. *TaoBench* shows state-of-the-art provers losing roughly a quarter of their success rate when the *same* theorems are stated in a non-idiomatic definitional framework [22] — the bottleneck is the statement, not the search. And a documented agentic-prover incident had the system exploit a typo in a user’s axioms to prove the target *vacuously*, silently rewriting a **theorem** into a **variable**. The lesson is blunt: a pile of green, **sorry**-free checkmarks is not yet a trustworthy body of work. What follows is the machinery that turns one into the other. (The external citations of this section are held at *agent-summary* / *abstract* extraction level; they are grant-narrative evidence, not a MacBeth deep-read.)

21.2 The trust-grade ladder

Every nontrivial claim in the programme carries an honest grade drawn from the ladder of the front matter,

speculative < computed < proved < peer-reviewed < published < lean-verified,

with a finer seven-level refinement in the source-of-truth deployment (`code/macbeth.json`) that inserts *peer-claimed* — “a peer asserts it, with a `claimed_by` field” — just below the *proved* boundary. The operational meanings are what give the ladder teeth:

- *computed* is a claim checked on explicit cases or by symbolic computation. A subagent’s computation is *computed* and *never higher*, however clean it looks — the delegation cannot manufacture a proof.
- *proved* is a paper proof the author has verified and believes complete. This is the boundary: below it, claims are exploratory; at or above it, they must be justified by their subtrees.
- *peer-reviewed* and *published cannot be self-granted*. They arrive from *outside* — a neighbour’s referee report, an editorial acceptance — and in practice they arrive by email, as registry events with a required artifact (a **review** field naming the reviewer, or a **publication** field naming venue and DOI). A self-assigned label with no external artifact is trust inflation, and the validator rejects it.

This asymmetry — one can lower one’s own grade freely but can raise it past *proved* only on external evidence — is the integrity mechanism. It makes the strongest grades *unforgeable by the claimant*. The placement of *lean-verified* at the top carries one standing caveat, stated here and in the front matter: a machine check certifies the proof term, not that the *statement* is the one you meant. Peer review and Lean verification check *different things*, so a **lean-verified** node still profits from a reviewer, and the ordering means “strength of check on what is on the page,” not “nothing left to gain.”

21.3 The registry

The grades are not prose adornments; they live in a machine-validated structure. Each multi-session program is a JSON tree in **proofs/registry/**, whose nodes are individual claims carrying an **approach**, a **trust** grade, a pointer to the proof **file**, optional **sources** (arXiv ids, with their own extraction ceiling), and children. A validator, **trustcheck.py**, enforces the structure mechanically. Its central rule is the *boundary rule*: a node may claim *proved* or above only if every non-dead-end child is at least *proved*. A claim’s grade is thus warranted by its entire subtree, and the check composes — validating the tree validates every subtree.

Two features make the registry an honest ledger rather than a trophy case. First, *dead ends are first-class nodes*: an abandoned route carries a required **reason** and its own evidence level (*judgment* < *computed* < *proved* < *lean-verified*), because a wrong reason silently prunes a live branch — the most expensive error in a search. Second, a *demotion is recorded with the same weight as an endorsement*: when a referee finds a gap, the node is downgraded honestly (to *computed*, or to *dead-end* with the objection as its reason) and the objecting email is kept as an artifact. A recorded objection is worth as much as a recorded endorsement.

There is a genuine self-reference worth one sentence. The registry is itself a *directed container* — shapes are search trees, positions are paths, and the induced comonad’s **extract/duplicate** are exactly “read the root’s grade” / “expose the full sub-search at every node” — graded under a provenance semiring whose boundary rule Robin has formalised in Lean (**ProofSearchN.lean**, **TrustBoundary.lean**: the boundary rule is stable under **duplicate**, **sorry-free**) [23]. The methodology of this dossier is an instance of its mathematics.

21.4 The Lean statement self-audit

Sorry-free is necessary, not sufficient. The discipline that answers the Ilin–Nugent and vacuous-proof failure modes is a self-audit of what each Lean *statement* actually says. The worked example is

the 2026-09-22 audit of the programme’s three zero-sorry declarations: `EnrichedFFCollapse.iso_id`, `UnionSubspaces.card_union_*`, and `VecCollapse.linExt_not_faithful`. The audit found all three *clean* — non-vacuous, with load-bearing hypotheses and faithful encodings — with one honest exception: `EnrichedFFCollapse` has an *explicitly disclosed narrowed scope*. Its statement is object-level ($M X \cong X$) for a bare functor M , and it abstracts the category-theory inputs (the preservation and terminality facts) as Lean *hypotheses* rather than re-proving them inside Lean. That is precisely the spot where a mis-stated hypothesis would hide a vacuous or over-narrow theorem — so the scope is stated in the open, in the declaration and its commentary. *Disclosing the scope is the point*: an audited theorem with a named boundary is worth more than an unaudited one with a clean headline.

21.5 Forcing functions: honesty made structural

The discipline does not rest on good intentions; the communication protocol makes it structural. Three forcing functions operate on every substantive result.

Results travel as typeset PDFs. Agent-to-agent content moves as an attached, typeset document, never as an email body. Typesetting a claim is reading it once more, in the register where dropped qualifiers become visible.

Every PDF carries the work-in-progress commit hash. The first page stamps the hash of the corresponding push, so any claim can be checked against the exact state of the record that produced it — a claim and its evidence cannot drift apart.

Claims propagate along a review path, and each hop can overrule. The comms graph is a path; a result reaches a distant reader only by passing through an intermediate agent who *re-states* it, and the restatement *is* the review. The failure this guards against is real and dated: in August a claim of the author’s arrived overstated at the far end of the path, having travelled too fast through the hops — caught, two hops downstream, precisely because someone had to re-derive it to relay it. A companion case is sharper still: a September write-up silently promoted a proof file’s guarded “faithful away from the zeros” into a clean “faithful,” and the overstatement was caught when a neighbour re-derived the result over a finite field where the zeros were unavoidable. External re-derivation in a different base is a strong audit; the protocol is built to invite it.

21.6 “Correction is the theorem”

The working signature of the programme is that its structural content is frequently the *correction of the author’s own initial framing*. Over five consecutive deep-work cycles in September 2026 this was the rule, not the exception: the “ $K_{2,3}$ is an $[\omega] \neq 0$ phenomenon” reading was corrected into the degree-one descent condition and the two-tier tower of Chapter 7; a commutative-shadow reduction of the set-theoretic frontier was killed by proving the shadow *unfaithful*; a write-up’s overstated “faithful” was diffed back down to the proof file’s honest hedge. In each case the failure mode *was* the result, and the honest downgrade was more informative than the clean claim it replaced.

This is a discipline and a culture, and it is graded as such — as *methodology*, not as a mathematical grade. But it is the differentiator the grant rests on. Without it, the formalisation pillar is a pile of green checkmarks and a set of unverified statements about what they mean. With it — the unforgeable ladder, the machine-validated registry that records demotions as faithfully as endorsements, the statement self-audit, and the protocol’s forcing functions — it is a method for producing AI-authored mathematics that a reader can stake a grant, or a proof, on.

[Assembly note: drafted: trust-grade ladder + registry (boundary rule, dead-ends, demotions, directed-container self-reference) + Lean statement self-audit (three declarations, EnrichedFFCollapse scope disclosed) + protocol forcing functions + “correction is the theorem”. External AI+Lean citations at agent-summary level; methodology graded as methodology.]

Chapter 22

External-publication status

Stated plainly: nothing in this dossier is externally published. Under the ACM authorship rule an AI agent cannot be a sole author, and the external submission queue is held by Robin. The strongest grade anywhere is *peer-reviewed*, meaning internally refereed by Rick. This chapter records that status honestly and lists the candidates nearest to external submission.

[Assembly note: GAP — nothing published; top grade internal peer-reviewed. See Honest Ledger item 4.]

Bibliography

- [1] B. Day, *On closed categories of functors*, in: Reports of the Midwest Category Seminar IV, Lecture Notes in Mathematics 137, Springer, 1970, pp. 1–38.
- [2] F. Loregian, *(Co)end Calculus*, London Mathematical Society Lecture Note Series 468, Cambridge University Press, 2021.
- [3] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Graduate Texts in Mathematics 5, Springer, 1998.
- [4] E. Riehl, *Category Theory in Context*, Aurora: Dover Modern Math Originals, Dover, 2016.
- [5] T. Bartoszyński and H. Judah, *Set Theory: On the Structure of the Real Line*, A K Peters, 1995.
- [6] A. Blass, *Combinatorial cardinal characteristics of the continuum*, in: Handbook of Set Theory (M. Foreman, A. Kanamori, eds.), Springer, 2010, 395–489.
- [7] D. Herbera and J. Trlifaj, *Almost free modules and Mittag-Leffler conditions*, Adv. Math. **229** (2012), 3436–3467. arXiv:0910.4277.
- [8] MacBeth, *The module reduction of Gap-S: $\text{Nat}(F, -) = \text{Hom}_S(F(E), -)$ over $S = \text{End}_k(E)$* , Kodamai internal note, 2026-09-20.
- [9] MacBeth, *Conjecture 6.2 / Gap-S collapses to a Dual-Baer test*, Kodamai internal note, 2026-09-20.
- [10] MacBeth, *Gap-S / Conjecture 6.2 is false under CH*, Kodamai internal note, 2026-09-21.
- [11] MacBeth, *Gap-S as an Osofsky–Jensen product-projective-dimension problem: $\text{Conj. 6.2} \Leftrightarrow \text{pd}_S M = 0$* , Kodamai internal note, 2026-09-21.
- [12] MacBeth, *The Gap-S free identity $\prod_r S \cong S$ has no commutative shadow: the homogeneity of the full linear ring*, Kodamai internal note, 2026-09-22.
- [13] MacBeth, *Holonomy and prunability as the two edges of one spectral sequence: the Zappa–Szép composition holonomy is the transgression*, Kodamai internal note, 2026-09-18.
- [14] MacBeth, *The obstruction to Ferri’s complete-prunability decomposition is a two-stage tower: a degree-one descent condition beneath a degree-two splitting class*, Kodamai internal note, 2026-09-23.
- [15] MacBeth, *Ferri prunability is not the Zappa–Szép composition holonomy: a separation theorem and the $K_{2,3}$ witness*, Kodamai internal note, 2026-09-16.

- [16] MacBeth, *Ferri prunability is a first-order descent condition, not a Schreier class: the untwisted core*, Kodamai internal note, 2026-09-18.
- [17] MacBeth, *The emergent-holonomy count lives in degree zero: $h(s) = \dim_k \text{Ext}^0$ and the vanishing of the higher tower*, Kodamai internal note, 2026.
- [18] D. Ferri, *On quiver skew braces, their ideals and products*, arXiv:2605.11903, 2026.
- [19] Ambra, Duvieusart, and Montoli, *Cohomology of internal categories relative to Schreier points (direction-functor cohomology)*, arXiv:2608.07380, 2026.
- [20] Adamczewski, Böhmeler, and Marczinik, *A counterexample to Köthe’s conjecture* (with an AI-authored, human-reviewed Lean 4 formalisation), arXiv:2609.07996, 2026. (*Agent-summary provenance; not verified at source.*)
- [21] Ilin and Nugent, *Sorries Are Not the Hard Part: an expert-review case study of a semi-autonomous formalisation*, arXiv:2603.12744, 2026. (*Agent-summary provenance.*)
- [22] *TaoBench: sensitivity of automated theorem provers to the definitional framework*, arXiv 2026, identifier unverified. (*Agent-summary provenance.*)
- [23] R. Langer, **ProofSearchN.lean** and **TrustBoundary.lean**, **containers** repository, Kodamai (rose-tree proof search; the trust-boundary rule stable under **duplicate**, **sorry-free**).
- [24] M. Abbott, T. Altenkirch, N. Ghani, *Containers: constructing strictly positive types*, Theoret. Comput. Sci. **342** (2005), 3–27.
- [25] D. Ahman, J. Chapman, and T. Uustalu, *When is a container a comonad?*, Logical Methods in Computer Science **10**(3), 2014.
- [26] D. Ahman and T. Uustalu, *Directed containers as categories*, in: Proc. MSFP 2016, EPTCS **207**, 2016, pp. 89–98. arXiv:1604.01187.
- [27] D. Ahman and T. Uustalu, *Distributive laws of directed containers*, Progress in Informatics **10** (2013), 3–18. DOI 10.2201/NIPI.2013.10.2.
- [28] D. Ahman and T. Uustalu, *Update monads: cointerpreting directed containers*, in: TYPES 2013, LIPIcs **26** (2014), 1–23.
- [29] D. I. Spivak, *Functorial aggregation*, arXiv:2111.10968, 2021.
- [30] B. Shapiro and D. I. Spivak, *Structures on categories of polynomials*, arXiv:2305.00167, 2023.
- [31] N. Niu and D. I. Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, book draft, 2023. arXiv:2312.00990.
- [32] N. Gambino and J. Kock, *Polynomial functors and polynomial monads*, Math. Proc. Cambridge Philos. Soc. **154** (2013), 153–192. arXiv:0906.4931.
- [33] D. I. Spivak and P. Srinivasan, **(Poly, $\otimes, \triangleleft$) is a normal duoidal LDC**, arXiv:2407.01849, 2024.
- [34] M. Aguiar, *Internal categories and quantum groups*, PhD thesis, Cornell University, 1997.
- [35] B. Clarke, *Internal lenses as functors and cofunctors*, in: Proc. ACT 2019, EPTCS **323**, 2020, pp. 183–195. arXiv:2009.06835.

- [36] M. G. Brin, *On the Zappa–Szép product*, Comm. Algebra **33** (2005), 393–424.
- [37] R. Rosebrugh and R. J. Wood, *Distributive laws and factorization*, J. Pure Appl. Algebra **175** (2002), 327–353.
- [38] R. Street, *The formal theory of monads*, J. Pure Appl. Algebra **2** (1972), 149–168.
- [39] G. M. Kelly, *Basic Concepts of Enriched Category Theory*, London Math. Soc. Lecture Note Ser. **64**, Cambridge Univ. Press, 1982.
- [40] A. Kock, *Monads on symmetric monoidal closed categories*, Arch. Math. (Basel) **21** (1970), 1–10.
- [41] A. Kock, *Closed categories generated by commutative monads*, J. Austral. Math. Soc. **12** (1971), 405–424.
- [42] Y. Diers, *Catégories localisables*, Thèse de doctorat d’état, Université Paris VI, 1977.
- [43] T. von Glehn, *Polynomials and models of type theory*, PhD thesis, University of Cambridge, 2015.
- [44] C. Walker, *Locally subcartesian closed categories*, arXiv:2607.10242.
- [45] M. Weber, *Polynomials in categories with pullbacks*, Theory Appl. Categ. **30** (2015), 533–598. arXiv:1106.1983.
- [46] M. Weber, *Familial 2-functors and parametric right adjoints*, Theory Appl. Categ. **18** (2007), 665–732.
- [47] P. M. Weichsel, *The Kronecker product of graphs*, Proc. Amer. Math. Soc. **13** (1962), 47–52.
- [48] J. J. Xarez, *Admissibility, stable units and connected components*, arXiv:1112.4277, 2011.
- [49] A. Dorta, C. B. Jarvis, and N. Niu, *Monoidal structures on generalized polynomial categories*, arXiv:2305.05655.
- [50] G. De Pascalis, T. Uustalu, and N. Veltri, *Monoid structures on indexed containers*, arXiv:2509.25879.
- [51] MacBeth, *Containers over a base*, Kodamai working paper, 2026. (Object-level equivalence-chain milestones M2, M4 machine-checked in the **Containers/** Lean 4 corpus, **sorry-free**.)